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Chronology, Continuity, and Hiatus: Radiocarbon Dating of Monumental Mudbrick Architecture in Peru's Chicama Valley

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ABSTRACT

Cultural sequences often rely on relative or typological chronological techniques, even as radiometric methods producing absolute ages become increasingly common. This reflects the perception that typological approaches are faster and sufficiently reliable, whereas radiometric precision demands greater time and financial investment. Here, we present a strategy for obtaining radiocarbon ages from monumental mudbrick structures (*huacas*), reporting 72 radiocarbon dates spanning 4000–600 CAL B.P. from 13 sites in Peru's Chicama Valley. By capitalizing on non-archaeological disturbances, we can efficiently produce AMS ¹⁴C determinations and evaluate the chronological assumptions of typologies. Results show that existing relative chronologies are broadly sound but poorly suited to resolving fine-grained diachronic change. Our results suggest monument building in Chicama occurred as relatively short pulses of intensive activity separated by long hiatuses. This study shows the potential of an underutilized material—mudbrick—to yield valuable chronological, ecological, and cultural information while minimally disturbing archaeological resources.

ARTICLE HISTORY

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KEYWORDS

radiocarbon ages; Bayesian modeling; Chicama Valley; Peru; North Coast; monumental architecture; adobe bricks

Introduction

The North Coast of Peru offers opportunities to study one of the longest documented sequences of cultural development in the ancient Americas (Bazán 2023; Bird 1948; Bird, Hyslop, and Skinner 1985; Chauchat 1975; Clément 2015; Dillehay 2017; Franco Jordán, Gálvez Mora, and Vásquez Sánchez 2003; Larco Hoyle 1938; Millaire et al. 2016; Toshihara 2002). Accurate, precise chronologies are essential to understanding how North Coast cultural traditions developed and engaged with one another, as well as interacted with exogenous social and environmental dynamics. However, many analyses have largely relied on typological dating, a relatively coarse chronological tool. Furthermore, typological frameworks that were initially designed as chronological tools have been reinforced as conceptual schema, occasionally with the unintended consequence of obscuring fine-scale cultural processes (Ramón Joffré 2005).

Radiocarbon dating provides an obvious means of improving temporal resolution and refining interpretations of diachronic change. Recent critiques based on reanalysis of radiometric data demonstrate that many assumptions about evenly-paced, linear sequences of cultural change are not strongly supported (Carmichael 2019; Contreras, Marsh, and Rademaker 2025; Gamboa and Lau 2022; Koons and Alex 2014; Pardo and Rucabado 2016; Ramón Joffré 2005). Despite their potential, radiocarbon dates are typically produced through excavation and analysis strategies that are costly. Limited resources and research priorities often result in only a few radiocarbon samples being dated to constrain the ages of important contexts; this contributes to a radiocarbon record which, inadvertently but in aggregate, is spatially and temporally uneven.

Here, we outline a strategy that capitalizes on exposures from looting, locally termed *huaqueo*, and agricultural development to opportunistically sample dateable materials from archaeological sites. This approach combines survey and documentation methods to rapidly and inexpensively collect materials from stratified construction sequences. We use a minimally-invasive sampling workflow that integrates sampling with photogrammetric recording, recovery of materials through flotation, and Bayesian analysis of calibrated AMS ¹⁴C dates. Specifically, we explore the potential of adobe bricks—an often-overlooked earthen construction material—to yield archives of dateable organic and paleoecological material.

This pilot study yielded several significant results. We present 72 dates collected by the Chicama Valley Environmental Archaeology Project (CHEAP) from 13 archaeological sites (Figure 1, Tables 1, 2), representing several cultural phases between approximately 4000 and 600 years calibrated before present (CAL B.P.). We provide summaries and images of the sampled sites and profiles in Supplemental Materials 1 and 2. Our sampling strategy targeted architectural features with clear stratigraphic and depositional relationships, enabling Bayesian models to refine chronologies. Adobe and similar earthen materials from North Coast contexts proved to be rich archives, yielding abundant material suitable for high-precision AMS ¹⁴C dating alongside other archaeological and paleoecological data.

Our results lead to provisional yet significant hypotheses regarding the tempo of monumental construction in the Chicama Valley. We find that previously published adobe typological sequences are broadly accurate, but their

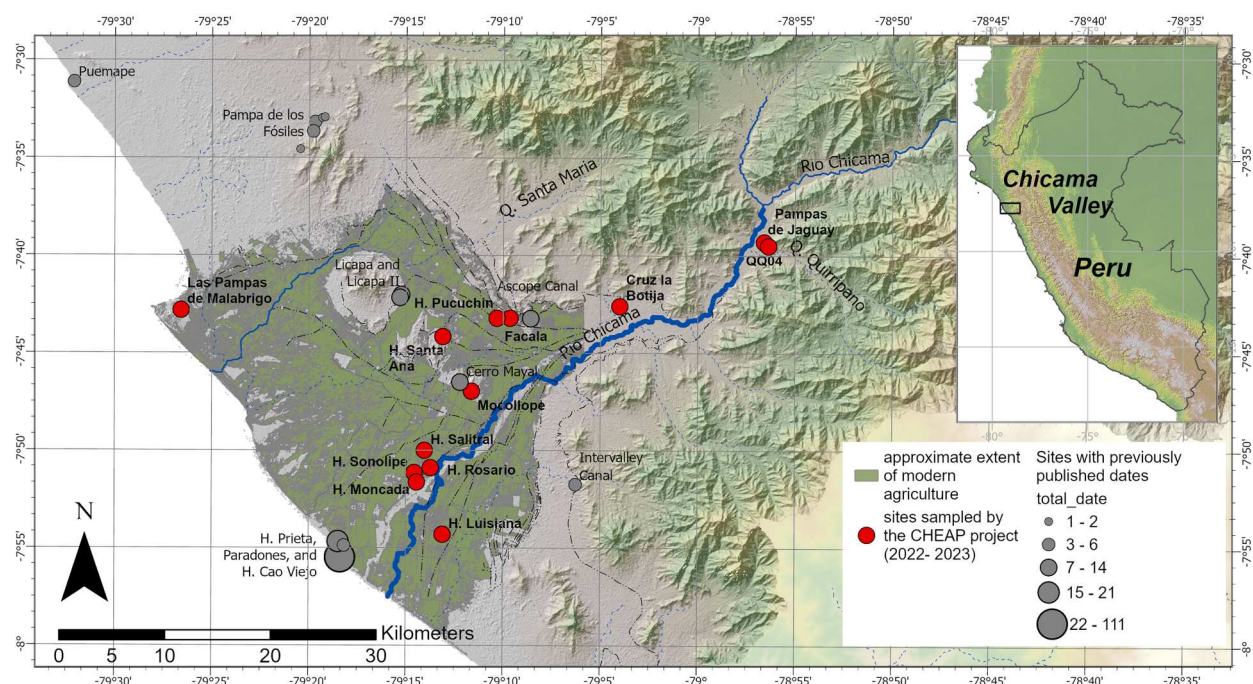


Figure 1. Map showing the location of sites in the Chicama Valley discussed in this article and sites for which radiocarbon dates have been previously published. Color versions of these figures are available online.

chronological sensitivity is often too limited to capture important cultural dynamics. Many chronological frameworks rely on a limited number of radiometric dates to anchor cultural sequences within broad periods of otherwise undifferentiated time, implicitly portraying change within periods as evenly paced. Rather than constant trajectories of monumental construction (and, presumably, cultural development), the age distributions for the events we sampled suggest monument building occurred during episodic pulses of intensive activity that were separated by longer apparent hiatuses. Construction techniques previously attributed to the Formative period dated as much as a

millennium earlier. Conversely, we found that age distributions of several types of building materials used during the 1st millennium A.D. partially overlap, potentially hinting at more complex social or technical decision making during monument construction than was suggested by prior sequences. These patterns were obscured by the resolution of prior chronological frameworks and were only resolvable by directly dating the adobes themselves. In short, the method we outline here is reliable and allows us to rapidly and efficiently establish valley-wide radiocarbon chronologies to address archaeological questions while opening avenues towards new research questions.

Table 1. Sites and contexts sampled for this analysis.

Site	UTM coordinates (Zone 17S)		Other designations from previous projects	Prior investigations	Sampled profiles (CHEAP)
Cruz la Botija	713233 E	9147285 S		Toshihara 2002	profiles 1–8
Castillo de Facalá	702829 E	9146178 S	REINDEL 42	Reindel 1993; Toshihara 2002; Ubbelohde-Doering 1952	profile 1–3
Púcuchin	700902 E	9147244 S	NA	NA	profile 1
Huaca Rosario	695249 E	9132081 S	Huaca Moncada (Ubbelohde-Doering 1952); PRACH 44; REINDEL 89	Larco Hoyle 1938; Ubbelohde-Doering 1952; Kosok 1965; Leonard and Russell 1992; Reindel 1993	profiles 1–2
Huaca Salitral	694750 E	9133696 S	Huaca de la Plaza/Huaca Plana (Leonard and Russell 1992); PRACH 20; REINDEL 90	Leonard and Russell 1992; Reindel 1993	profiles 1–2
Huaca Sonolipe	693725 E	9131613 S	Huaca del Palmo/Huaca Moncada (Ubbelohde-Doering 1952; Leonard and Russell 1992); PRACH 60; REINDEL 104	Larco Hoyle 1938; Ubbelohde-Doering 1952; Leonard and Russell 1992; Reindel 1993	profiles 1–2
Pampas de Jagüey	726880 E	9153287 S	CLEM-37	Gálvez Mora and Briceño 2001; Larco Hoyle 1946	profiles 1–7
Huaca Santa Ana	696502 E	9144424 S	Huaca Cerro Constancia, Huaca Garrapón (NASA-27)	Larco Hoyle 1945, 1948; Vining 2017	profile 1
Complejo Mocollope	699223 E	9139424 S	PRACH-128;	Attarian 2009; Franco Jordán 2016; Gálvez Mora, Becerra, and Marín 2002; Leonard and Russell 1992	profiles 1–5
Huaca Moncada	693970 E	9130745 S	Huaca Vallejos; PRACH-61	Leonard and Russell 1992;	profile 1
Huaca Luisiana	696434 E	9125783 S	NASA-32	Toshihara 2002; Vining 2017	profiles 1–4
QQ04	726574 E	9152611 S	NA	NA	
Las Pampas de Malabrigo	673450 E	9150285 S	NA	NA	

Table 2. Sample information and radiocarbon results. Lab codes: PSUAMS = Pennsylvania State University Radiocarbon Laboratory, DAMS = DirectAMS, OxAU = Oxford Radiocarbon Accelerator Unit, and UCIAMS = the University of California-Irvine Keck Carbon Cycle Accelerator Mass Spectrometer Facility. ^{14}C data are reported according to Stuiver and Polach (1977, 355–363) and Reimer, Brown, and Reimer (2004, 1299–1304). Results have been corrected for isotopic fractionation according to the conventions of Stuiver and Polach (1977), with $\delta^{13}\text{C}$ values measured on prepared graphite. Sample preparation backgrounds have been subtracted based on measurements of ^{14}C -free wood. All samples were pretreated in the University of Arkansas Environmental Archaeology Laboratory using a standard ABA procedure that follows UC-Irvine KCCAMS facility protocols (Stuiver and Polach 1977; Reimer, Brown, and Reimer 2004). These can differ from $\delta^{13}\text{C}$ of the original material, if fractionation occurred during sample graphitization or the AMS measurement and are not shown.

Lab number	Sample ID	Context	Material	F^{14}C	$1\sigma \pm$	D^{14}C (‰)	$\pm$	^{14}C age (B.P.)	$1\sigma \pm$	Est $\mu\text{g C}$
PSUAMS-16579	MC148 (bottle).	Pampas Jaguey, Profile 3 Context 2	Dry twig	0.8291	0.0019	-170.9	1.9	1505	20	566
PSUAMS-16580	MC149-FL049 (A).	Pampas Jaguey, Profile 3 Context 3	Dry twig	0.8335	0.0016	-166.5	1.6	1465	20	612
PSUAMS-16581	MC150-FL050 (A).	Pampas Jaguey, Profile 3 Context 4	Dry twig	0.8382	0.0016	-161.8	1.6	1420	20	616
PSUAMS-16582	MC154-FL054 (B).	Pampas Jaguey, Profile 4 Context 1	Charcoal	0.8035	0.0015	-196.5	1.5	1755	15	721
PSUAMS-16583	14C-119.	Pampas Jaguey, Profile 4 Context 2	Dry twig	0.8498	0.0016	-150.2	1.6	1305	15	671
PSUAMS-16584	14C-117.	Pampas Jaguey, Profile 4 Context 3	Dry monocot leaves	0.8036	0.0015	-196.4	1.5	1755	15	803
PSUAMS-16585	MC141-FL042 (B).	Pampas Jaguey, Profile 1 Context 1	Charcoal	0.8519	0.0015	-148.1	1.5	1285	15	732
PSUAMS-16586	MC152-FL052 (A).	Pampas Jaguey, Profile 5 Context 2	Dry twig	0.8483	0.0015	-151.7	1.5	1320	15	701
PSUAMS-16587	MC157-FL057 (A).	Pampas Jaguey, Profile 7 Context 3	Dry twig	0.8398	0.0016	-160.2	1.6	1405	20	604
PSUAMS-16588	MC135-FL036 (B).	Pucuchín, Profile 1 Context 2	Charcoal	0.6202	0.0013	-379.8	1.3	3840	20	778
PSUAMS-16589	MC138-FL039 (B).	Pucuchín, Profile 1 Context 4	Charcoal	0.6234	0.0014	-376.6	1.4	3795	20	653
PSUAMS-16590	MC134-FL035 (B).	Pucuchín, Profile 1 Context 1	Charcoal	0.6222	0.0014	-377.8	1.4	3810	20	724
PSUAMS-16591	MC137-FL038 (B).	Pucuchín, Profile 1 Context 5	Charcoal	0.6216	0.0015	-378.4	1.5	3820	20	581
PSUAMS-16598	MC158-FL058 (A).	QQ04, Profile 1 Context 1	Dry twig	0.7252	0.0017	-274.8	1.7	2580	20	448
PSUAMS-16599	MC158-FL058 (B).	QQ04, Profile 1 Context 1	Charcoal	0.6348	0.0015	-365.2	1.5	3650	20	737
PSUAMS-16600	14C-122 (B).	QQ04, Profile 2 Context 1	Charcoal	0.9029	0.0016	-97.1	1.6	820	15	693
PSUAMS-16601	14C-112 (C).	Facala, Profile 2 Context 1	Dry maize cob	0.8094	0.0016	-190.6	1.6	1700	20	716
PSUAMS-16602	MC130-FL031 (A).	Facala, Profile 2 Context 2	Dry twig	0.6893	0.0015	-310.7	1.5	2990	20	723
PSUAMS-16603	MC131-FL032 (A).	Facala, Profile 2 Context 3	Dry twig	0.8116	0.0015	-188.4	1.5	1675	20	681
PSUAMS-16604	MC100-FL001 (B).	Facala, Profile 1 Context 2	Charcoal	0.6270	0.0014	-373.0	1.4	3750	20	734
PSUAMS-16605	MC101-FL002 (A).	Facala, Profile 1 Context 4	Dry twig	0.8128	0.0015	-187.2	1.5	1665	15	696
PSUAMS-16606	MC102-FL003 (A).	Facala, Profile 1 Context 5	Dry twig	0.8095	0.0015	-190.5	1.5	1700	20	699
PSUAMS-16607	MC133-FL034 (B).	Facala, Profile 3 Context 1	Charcoal	0.8042	0.0015	-195.8	1.5	1750	15	675
PSUAMS-16608	MC129-FL030 (B).	Cruz La Botija, Profile 8 Context 1	Charcoal	0.6291	0.0014	-370.9	1.4	3725	20	680
PSUAMS-16609	MC128-FL029 (B).	Cruz La Botija, Profile 8 Context 2	Charcoal	0.6271	0.0014	-372.9	1.4	3750	20	719
PSUAMS-16610	MC103-FL004 (B).	Cruz La Botija, Profile 2 Context 1	Charcoal	0.6290	0.0016	-371.0	1.6	3725	25	508
PSUAMS-16611	MC168-FL068 (B).	Cruz La Botija, Profile 6 Context 5	Charcoal	0.6295	0.0015	-370.5	1.5	3720	20	714
PSUAMS-16612	MC172-FL072 (A).	Cruz La Botija, Profile 6 Context 6	Dry twig	0.6293	0.0014	-370.7	1.4	3720	20	649
PSUAMS-16613	14C-III (A).	Cruz La Botija, Profile 6 Context 10	Dry twig	0.6313	0.0014	-368.7	1.4	3695	20	691
PSUAMS-16614	MC120-FL021 (B).	Cruz La Botija, Profile 7 Context 6	Charcoal	0.6328	0.0014	-367.2	1.4	3675	20	696
PSUAMS-16615	MC124-FL025 (B).	Cruz La Botija, Profile 7 Context 5	Charcoal	0.6304	0.0014	-369.6	1.4	3705	20	761
PSUAMS-16616	MT474.	Pampas, Area 6 Quad 2e, Level sitting on 5	Squash skin	0.7900	0.0015	-210.0	1.5	1895	20	696
PSUAMS-16617	MT112.	Pampas, Area 6 Quad 2c, Level 3	Seed skin	0.7810	0.0016	-219.0	1.6	1985	20	608
PSUAMS-16618	MT33.	Pampas, Area 6 Quad 2c, Level 2	Seed skin	0.7963	0.0015	-203.7	1.5	1830	15	743
PSUAMS-16619	MC136-FL037 (B).	Pucuchín, Profile 1 Context 3	Charcoal	0.6277	0.0015	-372.3	1.5	3740	20	763
PSUAMS-17338	14C009	Huaca Rosario 001, Profile 2, Context 2	Poaceae cane	0.9003	0.0017	-99.7	1.7	845	20	530

(Continued)

Table 2. Continued.

Lab number	Sample ID	Context	Material	F ¹⁴ C	1σ ±	D ¹⁴ C (‰)	±	¹⁴ C age (B.P.)	1σ ±	Est μg C
PSUAMS-17339	14C034	Mocollope 004, Profile 1, Context 1	Peanut shell	0.7653	0.0019	-234.7	1.9	2150	20	392
PSUAMS-17340	14C072	Mocollope 004, Profile 3, Context 5	Charred twig	0.7958	0.0017	-204.2	1.7	1835	20	610
PSUAMS-17341	14C075	Mocollope 004, Profile 3, Context 12	Twig	0.8028	0.0016	-197.2	1.6	1765	20	658
PSUAMS-17342	14C051	Huaca Luisiana 005, Profile 2, Context 2	Charred twig	0.6266	0.0017	-373.4	1.7	3755	25	606
D-AMS-050786	14C001A	Huaca Rosario, Profile 1 Context 2	wood and grass	0.9064	0.0028			789	25	
D-AMS-050787	14C005A	Huaca Rosario, Profile 2 Context 1	Woody stems with leaves	0.9063	0.0029			790	26	
D-AMS-050788	14C005B	Huaca Rosario, Profile 2 Context 1	Charcoal	0.8407	0.0025			1394	24	
D-AMS-050789	14C008	Huaca Rosario, Profile 2 Context 2	Woody stems with leaves	0.9041	0.0023			810	20	
D-AMS-050790	14C018A	Huaca Sonolipe, Profile 2 Context 3	Wood and grass	0.9057	0.0025			796	22	
D-AMS-050791	14C020A	Huaca Salitral, Profile 1 Context 1	Grass	1.0112	0.0023			Modern		
D-AMS-050792	14C022A	Huaca Salitral, Profile 1 Context 5	Woody stems with leaves	0.9395	0.0024			501	21	
D-AMS-050793	14C029A	Mocollope, Profile 1 Context 2	Wood, grass, Seed pod	0.9293	0.0022			589	19	
D-AMS-050794	14C031A	Mocollope, Profile 1 Context 4	Grass	0.8331	0.0031			1467	30	
D-AMS-050795	14C033A	Mocollope, Profile 1 Context 6	Wood, grass, leaves	0.8334	0.0023			1464	22	
D-AMS-050796	14C053A	Huaca Moncada, Profile 1 Context 1	Wood, grass, seed pod	0.8901	0.0025			935	23	
D-AMS-050797	14C054A	Huaca Moncada, Profile 1 Context 2	Wood and grass	0.8952	0.0024			889	22	
D-AMS-050798	14C059B	Mocollope, Profile 2 Context 7	Charcoal and uncarbonized wood	0.8284	0.0022			1512	21	
D-AMS-050799	14C091A	Huaca Santa Ana, Profile 1 Context 3	Wood and grass	0.824	0.0023			1555	22	
D-AMS-050800	14C094A	Huaca Santa Ana, Profile 1 Context 6	Wood and grass	0.8231	0.0023			1564	22	
D-AMS-050801	14C095A	Huaca Santa Ana, Profile 1 Context 5	Wood and grass	0.8148	0.0024			1645	24	
D-AMS-050802	14C010	Huaca Sonolipe, Profile 1 Context 1	Seed pods	0.9031	0.0022			819	20	
D-AMS-050803	14C013B	Huaca Sonolipe, Profile 1 Context 3	Charcoal	0.9006	0.0022			841	20	
D-AMS-050804	14C016	Huaca Sonolipe, Profile 1 Context 5	Wood and grass	0.8951	0.0023			890	21	
D-AMS-050805	14C025B	Huaca Salitral, Profile 1 Context 7	Charcoal	0.8903	0.0021			933	19	
D-AMS-050806	14C036	Huaca Luisiana, Profile 1 Context 2	Charcoal	0.6351	0.0019			3647	24	
D-AMS-050807	14C038B	Huaca Luisiana, Profile 1 Context 4	Charcoal	0.6319	0.0018			3687	23	
D-AMS-050808	14C041B	Huaca Luisiana, Profile 3 Context 1	Charcoal	0.6334	0.0021			3668	27	
D-AMS-050809	14C043B	Huaca Luisiana, Profile 3 Context 4	Charcoal	0.6336	0.0021			3666	27	
D-AMS-050810	14C044B	Huaca Luisiana, Profile 3 Context 5	Charcoal	0.6316	0.0019			3691	24	
D-AMS-050811	14C045B	Huaca Luisiana, Profile 3 Context 6	Charcoal	0.63	0.0023			3712	29	
D-AMS-050812	14C046B	Huaca Luisiana, Profile 3 Context 7	Charcoal	0.6329	0.002			3675	25	
D-AMS-050813	14C052	Huaca Luisiana, Profile 4 Context 1	Charcoal	0.6308	0.0026			3701	33	
OxA 6896	RC-04	Cao Viejo, adobe from behind collapsed wall	<i>Phragmites comunis</i>					1480	40	
UCIAMS-277382	108	Cao Viejo, decorated wall on last south patio, coord. Y:45-48 R:297-300	Botanical	0.8173	0.0013	-182.7	1.3	1620	15	
UCIAMS-256455	1000-44	Cao Viejo, unit 2, A4, A5, B4, B5, UE21 Marked adobe	Botanical	0.8174	0.0014	-182.6	1.4	1620	15	
UCIAMS-256464	1000-116	Cao Viejo, Poligono 1, cuads. A3, A4, B3, B4, UE14	Botanical	0.8205	0.0013	-179.5	1.3	1590	15	
UCIAMS-277380	100	Cao Viejo, south wall	Botanical	0.8158	0.0014	-184.2	1.4	1635	15	

Background

Challenges and potentials for establishing chronology in deeply stratified sites

Our research area is the coastal valley of the Chicama River, in northern Peru (see [Figure 1](#)). Many archaeological loci in Chicama are large, monumental earthen structures—locally termed *huacas*. Huacas were built through successive episodes of construction, renovation, and reutilization, resulting in complicated but stratigraphically-organized sequences of events. As such, huacas share commonalities with some mounded sites found globally, and the methods we outline here could be adapted as appropriate. Three characteristics are pivotal to huacas' chronological interpretation: 1) huacas were built in discrete construction episodes rather than through accretional processes, and 2) the organic materials incorporated in adobe can provide a direct chronometer of these episodes, but 3) only if it can be reliably demonstrated that neither the organic material nor the event of adobe manufacture itself greatly predate the construction episode.

Deeply stratified sites have great archaeological potential but demand significant commitments of time and resources to properly and ethically excavate and document. Furthermore, large-scale excavations have lost favor over recent decades in favor of targeted and problem-oriented research which conserves finite archaeological resources (unless mitigating circumstances like triage of threatened sites warrant larger interventions). These factors often discourage projects from pursuing deep excavations in monumental huacas that might facilitate building long chronological sequences. Chapdelaine highlighted this challenge with respect to Moche huacas: “Moche civilization is a long cultural phenomenon ... [whose] beginnings are not well established through relative and radiometric dating, *mostly because layers pertaining to these earlier times are buried under meters of refuse*” (Chapdelaine 2011, 195, emphasis added). Koons and colleagues (2024, 83) note a similar “relative lack of data from earlier Moche contexts” in their analyses of ceramic chronologies. More recent projects have addressed the chronology of Moche monuments and, in the case of Huaca Prieta, very early cultural phases through deep excavations (Dillehay et al. 2012; Koons 2015; Uceda et al. 2021). But, robust control over the chronology of North Coast monument building is still wanting for much of the region's prehistory.

A net, unintended consequence of all these circumstances is that chronologies remain poorly resolved, making it difficult to address key questions, such as: what was the timing and tempo of construction (punctuated or continuous, across and within periods), at diverse sites, including huacas, agricultural infrastructure, and smaller sites? How did these developments within Chicama articulate with cultural, demographic, or environmental developments in adjacent coastal valleys and highlands? Adobes—unfired earthen bricks—and similar earthen materials provide opportunities to address these and other questions. Adobe appears early on the Peruvian coast (Fuchs et al. 2009; cf. Mauricio et al. 2021), and variations in form have been argued to be temporally sensitive. Their assumed calendrical ages have rarely been radiometrically tested, however, even while the bricks provide opportunities to do so. Earthen construction

materials incorporate carbonized and uncarbonized organic materials within their matrices, yielding the requisite source material for radiocarbon dates.

Few other studies have explored the potential of adobe and similar mudbricks to establish absolute chronology. Dee and colleagues (2013) dated vegetal material in Egyptian mudbrick but found resulting ages were consistently older than expected; they attribute this to old carbon in the alluvial sediments used to manufacture the brick. In contrast, Unkel and colleagues (2007) derived reliable AMS ^{14}C dates from southern Peruvian adobe. Similarly, Feathers, Quilter, and LeBlanc (2024) published single-grain luminescence dates from adobe from Moche sites in Chicama and elsewhere. Recent papers further demonstrate that material incorporated within earthen bricks offers a potential wealth of environmental proxy data (Alvaro Quispe 2020; Arbøll et al. 2023; Patalano et al. 2022). As earthen building materials represent well-defined, cohesive sedimentary packets, these proxies can be securely tied to precise chronologies. Given their ubiquity and this potential, adobe is underutilized as an important archive of archaeological information.

The majority of North Coast archaeological sites include at least some adobe construction and exhibit some degree of destruction from illicit excavation or agricultural and/or residential development (Vining 2017). While unequivocally lamentable, the silver lining is that such destruction cuts into deeply stratified huacas, exposing otherwise inaccessible architectural sequences. Construction events can be documented and sampled opportunistically without large-scale excavation. The abundance of adobe bricks from successive building episodes exposed in these disturbances offers enormous potential to develop stratigraphic sequences of absolute ages and Bayesian age models.

Previous chronological work in the Chicama Valley

The North Coast chronologies upon which archaeologists rely principally derive from seriated and stratigraphic dating techniques. These chronologies reflect the mid-20th century A.D. methods available when they were first proposed but later became entrenched as conceptual models explaining sequences of cultural change (Ramón Joffré 2005). Several typologies, such as Larco Hoyle's Moche sequence (1946, 1948), were based on collections that lacked significant contextual data by many current standards, making it a challenge to parse coeval or sequential relationships. Even critiques of models like Larco's are rarely grounded in radiometric determinations (e.g., Castillo and Donnan 1994). Recent reassessments which pair typological criteria with radiometric data (Cleland and Shimada 1992; Koons and Alex 2014) have greatly improved chronological control, while highlighting some of the limitations of prior sequences based solely on stratigraphic, typological, or seriation criteria.

Despite their obvious potential, absolute dating methods have played a small role relative to the extensive archaeological research conducted on the Peruvian North Coast. Approximately 250 archaeological radiocarbon dates for the Chicama Valley are previously published (Supplemental Material 3). Although substantial, the resulting radiocarbon record is unevenly distributed in space and time (see [Figures 1, 2](#)). The majority of dates comes from fewer than 10 well-

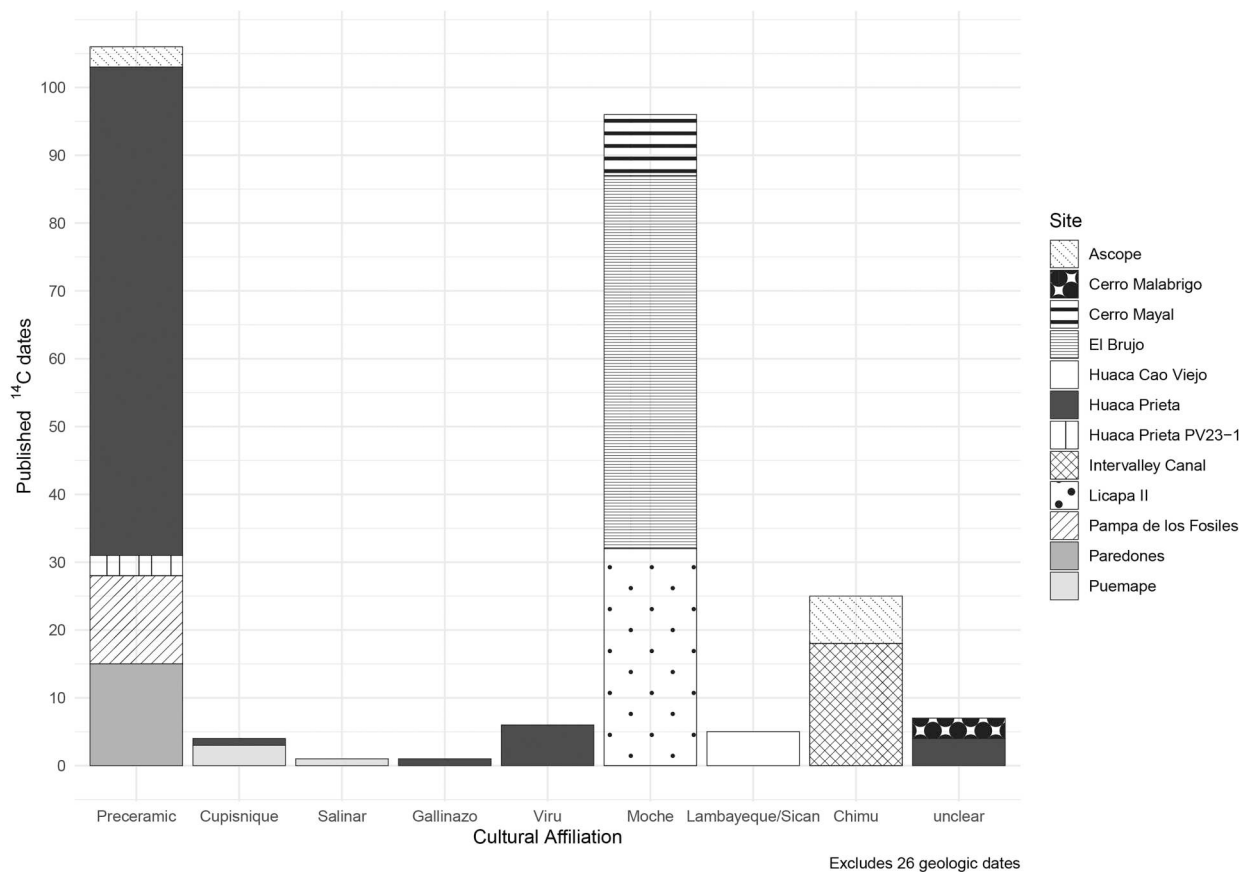


Figure 2. Bar plot showing the frequency of previously published radiocarbon dates from each site per period.

investigated sites, with > 60% coming from just three (see Figure 1). Most are from monumental contexts at el Brujo Archaeological Complex (a Pleistocene geological terrace where several sites were built); few other site types or even huacas elsewhere in Chicama have been radiometrically dated. Huaca Prieta and Sector Paredones at el Brujo provide the majority of Archaic–Formative period dates (Bird, Hyslop, and Skinner 1985; Dillehay 2017; Dillehay et al. 2012; Millaire et al. 2016), with additional dates from Pampa de los Fosiles and Puémape (Elera Arévalo 1998, table 119; Goldberg, Mychajliw, and Hadly 2016). Excavations at Huaca Cao Viejo and nearby Magdalena de Cao Viejo, and also at el Brujo, provide radiocarbon dates for the Moche, Transitional, and Colonial periods (Bazán 2022; Franco Jordán, Gálvez Mora, and Vásquez Sánchez 2003; Quilter 2011). Moche phases have also been dated at Licapa II, Cerro Mayal, and Mocollope (Attarian 2003; Koons et al. 2024; Russell et al. 1994). Lambayeque-period dates in Chicama are limited to a single large cemetery at el Brujo (Oakland 1999; Santana-Sagredo et al. 2020). Currently, the chronology of Chimú sites in the Chicama Valley is constrained by dates on two agricultural systems (Huckleberry, Caramanica, and Quilter 2017; Kus 1972; Pozorski 1987; Tavera-Medina et al. 2025).

The resulting radiocarbon record is uneven, potentially reflecting dating opportunities rather than cultural dynamics. There are significant gaps for the periods after ca. 3000 B.C. and for certain site types (see Figures 1, 2). Ceramic typologies continue to be the primary means of establishing chronological control. However, these are limited by coarse temporal resolution, and many cultural dynamics remain inaccessible. The dates we report here increase the

total sample of radiocarbon ages from Chicama by approximately 22% and more than double the number of sites represented, contributing key data for previously underrepresented periods.

Construction materials as chronologically sensitive artifacts

North Coast huacas exhibit a variety of construction materials and techniques. Since approximately 2000 B.C., the most widespread have used various forms of mudbricks (*adobes*) and different methods for laying them (bonding pattern). Other indigenous construction techniques employed stone masonry, compressed or pounded earth (*tapia*), and a technique (*quincha*) of embedding flexible cane within soft mud, similar to wattle and daub. The earliest currently known use of adobe-type construction materials, radiometrically dated to ca. 5100 CAL B.P., occurs at los Morteros in the Chao Valley, approximately 98 km to the south (Mauricio et al. 2021). Formally-shaped adobe bricks became widespread at a later point (Fuchs et al. 2009).

Functional explanations (e.g., Brooks et al. 2005; Shimada and Cavallaro 1985) have been proposed to explain choices between building materials, their shapes, and their use. Gálvez Mora and Castañeda Murga (2014, 400) propose that variable types reflect technical requirements, with some forms manufactured by specialists and others by non-specialists lacking specialized knowledge or equipment. For example, they interpret mold-made rectangular-convex adobes at Chimú sites in Chicama as the product of “experienced bricklayers,” while hypothesizing that irregular, hand-formed types (e.g., plano-convex) were contributed by

unskilled laborers to meet escalated labor demands (2014, 401).

Adobe form is most often presumed to reflect cultural choices and, as such, changes in form have chronological importance. Implicit in this argument is the notion that adobe form is sensitive to socially-structured technological knowledge and norms and can be used to identify constructions built by different North Coast groups. This argument follows a “community of practice” logic (Roddick 2016; Wenger 1999), where shared social identities emerge among individuals who invest collaborative effort in long-term and practical cooperative projects. As such, various ways of manufacturing adobes and monuments from them may reflect interactions among social groups. Adobe as the material vestige of communities of practice opens up many avenues into trying to understand the social landscape of North Coast monumentality but will require dedicated research into patterning of the variability in adobe technology and form that we begin to document here.

There is significant interest in using chronostratigraphy and seriation to link particular types to cultural periods (Campana Delgado 2000; Cavallaro and Shimada 1988; Hastings and Moseley 1975; Kolata 1982; Larco Hoyle 1946; see various in Millaire 2009; Reindel 1993, 1997; Shimada and Cavallaro 1985; Toshihara 2002; Tsai 2012; Tschauner 2001; Ubbelohde-Doering 1952). Reindel, using adobe form, brick bond (stacking pattern), and related variables from 123 Formative–Late Horizon (ca. 800 B.C.–A.D. 1530) huacas in six North Coast valleys, provides the most comprehensive seriation (1993, 1997). Campana Delgado (2000) proposes an explicitly evolutionary framework, arguing that adobe types track lines of cultural inheritance in addition to chronology. Gálvez Mora (2012) provides a typology specific to the Chicama Valley, identifying six types for the Middle–Late Horizon periods. Others (Shimada 1990; Shimada and Cavallaro 1985; Tsai 2012; Tschauner 2001) propose relative chronologies using manufacturer marks alongside morphology and construction techniques. However, manufacturer marks are primarily attributed to the social organization of labor groups (Cavallaro and Shimada 1988; Hastings and Moseley 1975; Moubarac 2002; Shimada and Cavallaro 1985; Tsai 2012).

Conical adobes and cobble-based construction on the North Coast are commonly attributed to the Cupisnique and other Formative-period cultures (Bird 1948; Bird, Hyslop, and Skinner 1985; Larco Hoyle 1938; Reindel 1993). Cane-impressed adobes have variably been attributed to Virú, Gallinazo, and Moche cultural traditions (Bennett 1950; Donnan 1973; Gálvez Mora 2012; Hastings and Moseley 1975; Millaire and Morlion 2009; Shimada 1994; Shimada and Maguiña 1994; Strong and Evans 1952; Wilson 1988). At Huaca del Sol and Huaca de la Luna in the Moche Valley, cane-impressed adobes occur most frequently (8–27%) in the earliest phases of each huaca but subsequently decline to <5% (Hastings and Moseley 1975, table 1; Moseley 1975). A similar transition occurs at Huaca Cao Viejo, where lower strata with plano-convex and cane-impressed adobes are replaced in upper levels by thicker, square adobes with smoothed sides and maker’s marks (Bazán and Alva 2024; Gálvez Mora et al. 2003). Kolata proposed a sequence of Chimu rectangular cuboid (or rectangular parallelepiped) bricks at nearby Chan Chan that varied based on their height-width ratios (Kolata 1978, 1982; cf. Narvaez 1989).

Despite the interest in developing and applying these typologies as chronological tools, their chronological assumptions are largely untested against radiometric data.

Description of sampled archaeological sites

We sampled 39 profiles at 13 sites in the Chicama Valley, developing absolute chronologies for construction events that looting and agriculture had exposed (see Figure 1). These contexts span the late Archaic transition through the Chimú period, approximately 2000 B.C.–A.D. 1400. Several sites were previously described by projects that did not provide absolute ages, while others were not previously documented. Brief descriptions of these sites, prior research, and information on the sampled contexts are provided in Supplemental Material 1, and orthophoto models and drawings are provided in Supplemental Material 2.

Methods

Locus selection process

Partial destruction of archaeological sites due to illicit excavations or agricultural and infrastructure development is ubiquitous on Peru’s North Coast, and it is extremely rare to encounter a site without disturbance. Shallow looting exposes near-surface architecture and provides opportunities to establish terminus ante quem (TAQ) dates for underlying architectural phases. Industrial-scale sugarcane cultivation is pervasive in Chicama, posing another threat to sites (Vining 2017). The margins of many huacas have been pared away by year-over-year deep plowing during sugarcane planting. This exposes ultimate, penultimate, and occasionally earlier construction events towards the exterior of the huacas’ monumental edifices. In extreme scenarios, looting trenches and tunnels may cut deeply into the center of sites. In the most severe instances, large machinery has cut trenches deeply into or even through sites. These are opportunities to collect TAQ and terminus post quem (TPQ) materials for much of a structure’s building sequence, even if the earliest events cannot be established. Feathers, Quilter, and LeBlanc (2024) capitalized on one such trench at the aptly named Huaca Cortada. We sampled a similar cut at Huaca Moncada.

We identified disturbed contexts by comparing historical *Servicio Aerofotográfico Nacional de Peru* (SAN) imagery acquired in 1943 to more recent IKONOS (acquired April 22, 2002), Worldview 2 (acquired Sep 13, 2013), and Planet labs (acquired 2022–2023) multispectral images. Looting is readily apparent in the imagery. This image series also helped identify sites whose footprint has been reduced over the past 80 years, exposing parts of their edifices. Estimates of percent change between 1943 and 2023 were compared against prior survey results to target cultural contexts for which we wished to improve chronological control. We visited these locations in the field to verify that suitable exposures were present and that identifiable adobe forms and sequences of construction were exposed.

Sampling strategies

We targeted sites which we estimated were built between approximately 3000 B.C. and A.D. 1500 and which comprised a transect across the lower valley to expand the available

sample beyond the clusters of locations that were previously sampled. Our principal objectives were to establish radiometric dates for construction phases and identifiable brick types and to assess whether the same contexts could yield productive paleoenvironmental proxies. We collected complete and fragmented adobe, other construction materials, and subfossil macroscopic botanical remains and marine bivalve shells (*Donax* sp. and *Mesodesma* sp.) for schlerochronological analysis (Table 3). We prioritized sampling at sites where at least two profiles were exposed, each showing minimally two and ideally three or more distinct construction episodes. We identified building phases from changes in construction materials or styles, plastered or faced exterior surfaces covered by later structures, or architectural elements (e.g., doorways or rooms) infilled by post-abandonment or closure deposits. When these criteria were met, the resulting dates could be used in a Bayesian analysis of the phases' sequence. Under ideal circumstances, these profiles would reveal events relatively early and late in the structure's construction sequences. Although it is difficult—if not impossible—to determine if these were really initial and terminal construction events, they provide TPQ and TAQ dates for other construction episodes.

Once identified, profiles were minimally cleaned, removing only loose sediments that had accumulated from looting or aeolian/colluvial activity after the site had been disturbed. We documented construction sequences using modified Harris matrices to record the relative stratigraphic position of each construction episode.

Site and profile documentation

We recorded each structure and sampled context with photogrammetry. Using a DJI Phantom 4 UAV with a SONY ILCE 20 mp CMOS sensor, we took low-altitude photographs of each site, overlapping frames by ca. 80% to create true-color and 2.5D digital terrain models (DTM). We established ground-control using survey targets that were visible in each photomosaic. Additionally, we created photogrammetric models for each profile after cleaning and prior to sampling. Profiles were photographed using a Canon PowerShot S110 (12.1 MP 1/1.7 CMOS sensor) camera. Samples were coded/tagged and photographed while in situ. Minimally five AgiSoft photogrammetry coded targets were placed at the corners and centers of each exposure for

spatial control. We generated both profile and site orthographic photomosaics and DTM in AgiSoft Metashape (v. 1.5.5). The locations of all targets and each collected construction, macrobotanical, and malacological sample recorded using an EMLID Reach RS2 RTK-GNSS receiver paired with a local base. Mean accuracy for each XYZ position was approximately 2 cm.

Conventional recording methods (field forms and profile drawings) were highly valuable even with current photogrammetric technologies—neither serves as a complete substitute for the other. We sketched profiles, noting the visible construction episodes, the locations of materials we selected for sampling, and other pertinent information with forms and sketches before ultimately converting them to Harris matrices. This documentation served as a scaffolding for converting the digital workflow to vectorized and annotated graphical records of the exposures and samples (Figure 3). Once recorded, samples were pulled from the profile, separately labeled, and bagged. When we sampled complete adobes, their size and type was recorded prior to extracting ^{14}C and other materials. Sizes and shapes of adobes left in situ were also recorded from photogrammetric models.

Recovery of ^{14}C and paleoecological materials through flotation

We recovered the majority of radiocarbon and paleoecological samples by disaggregating adobe brick and mortar in a flotation system (Figure 4) modified from the design published by Hastorf (1999). The principal modifications entailed scaling up components to suit the greater volume, mass, and induration of construction materials. It was critical to percolate water through the samples using a perforated dispersal head to deflocculate the indurated mud. Other modifications included: a more robust 0.5 cm mesh made with 0.063 gauge wire to support the heavier bricks; the addition of a second reservoir and return pipes to create a recirculating system; a stronger 1 hp water pump with two ball valves to regulate flow and percolation pressures; an auxiliary hose to aid in cleanup and dissolving samples; and, a large clean-out valve at the bottom of the primary circulation tank to remove the muddy sediments that quickly accumulated (see Figure 4).

Prior to processing, bricks were photographed, measured, weighed, and a 0.05–0.5 kg sample removed for

Table 3. Summaries of principal materials collected from adobe bricks via flotation.

	Flotation sampling						Field sampling	
	Marine shell		Carbonized wood		Uncarbonized botanical remains		Marine shell	
	Count	Total weight (g)	Count	Total weight (g)	Count	Total weight (g)	Count	Total weight (g)
Huaca Luisiana	15	17.2	15	33.5	14	33.4	7	10.1
Cruz la Botija	13	4.5	38	37.2	9	12.1	NA	NA
Pucuchin	6	6.3	7	13.6	-	-	NA	NA
Facala	3	0.8	2	1.2	7	20.7	NA	NA
QQ04	-	-	1	1	1	1	NA	NA
Las Pampas de Malabrigo	-	-	8	1.9	12	37.8	NA	NA
Pampas de Jaguey	1	0.4	12	1.9	8	28.7	NA	NA
Huaca Santa Ana	5	0.2	5	4.5	5	4.5	-	-
Mocollope	10	23.9	10	23.8	10	23.9	19	7
Huaca Cao	NA	NA	NA	NA	NA	NA	NA	NA
Huaca Moncada	2	0.6	2	8.4	2	8.4	9	6.02
Huaca Rosario	7	3.5	-	-	7	10.1	1	0.12
Huaca Salitral	7	9.4	8	3.9	11	4.1	16	9.01
Huaca Sonolipe	10	5.7	10	0.5	24	22.6		

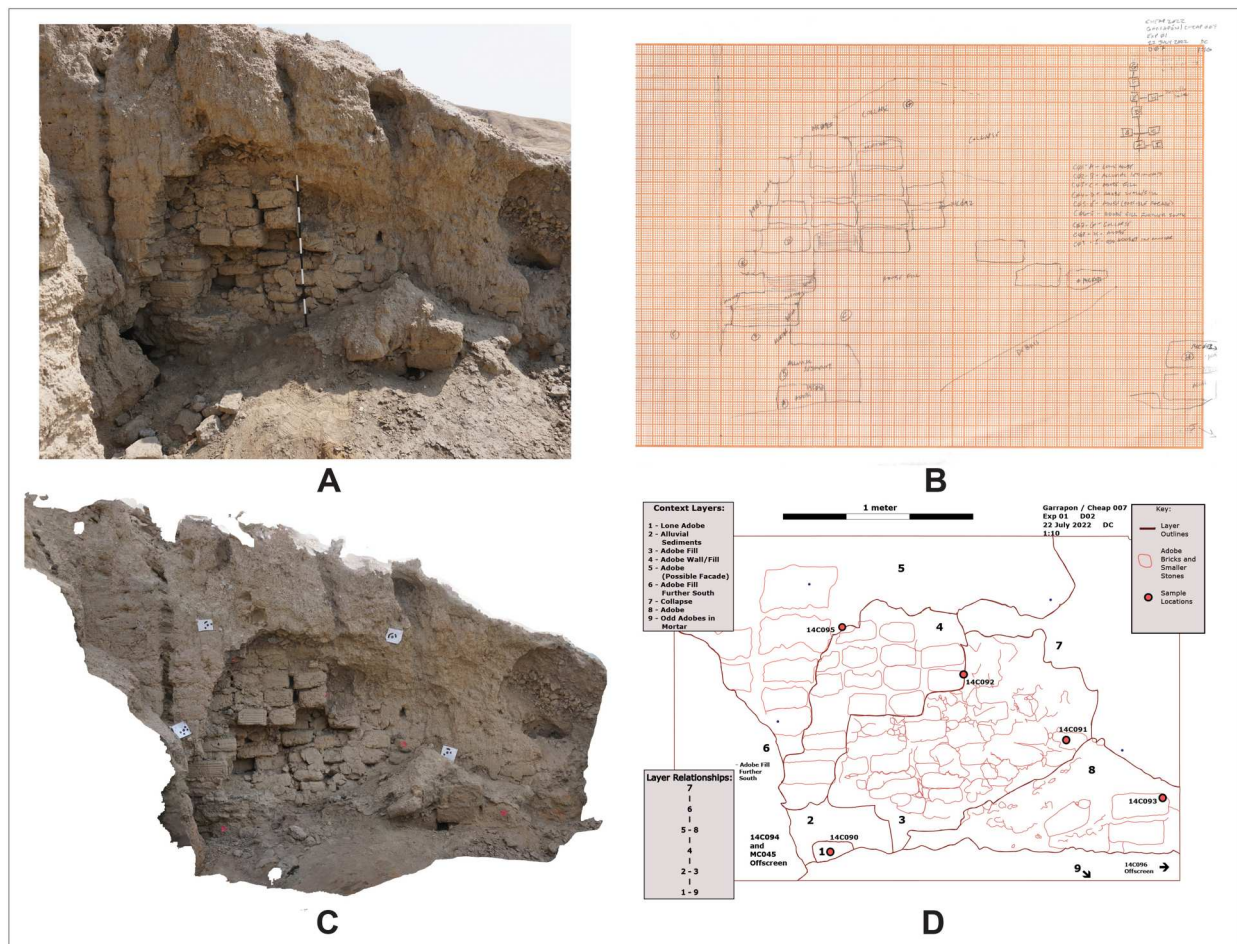


Figure 3. Schematic diagram showing the process for generating 2.5D photogrammetric models from the exposed profiles.

archiving. We manually cleaned the surfaces of each sample to remove any foreign or intrusive material, such as rootlets, that had attached while the profile was exposed. The remaining sample was subsequently dissolved. Light fraction was separated by a decanting spout overflowing into a 0.5 mm nylon mesh placed over the recirculation tank

(see Figure 4). Heavy fraction with > 0.5 cm diameter was captured by the heavy mesh. These were primarily pebbles incorporated into adobe, but malacological and ceramic fragments were also recovered in this way (see Table 3). The heavy fraction with < 0.5 cm in diameter settled to the bottom of the primary flotation tank and was flushed

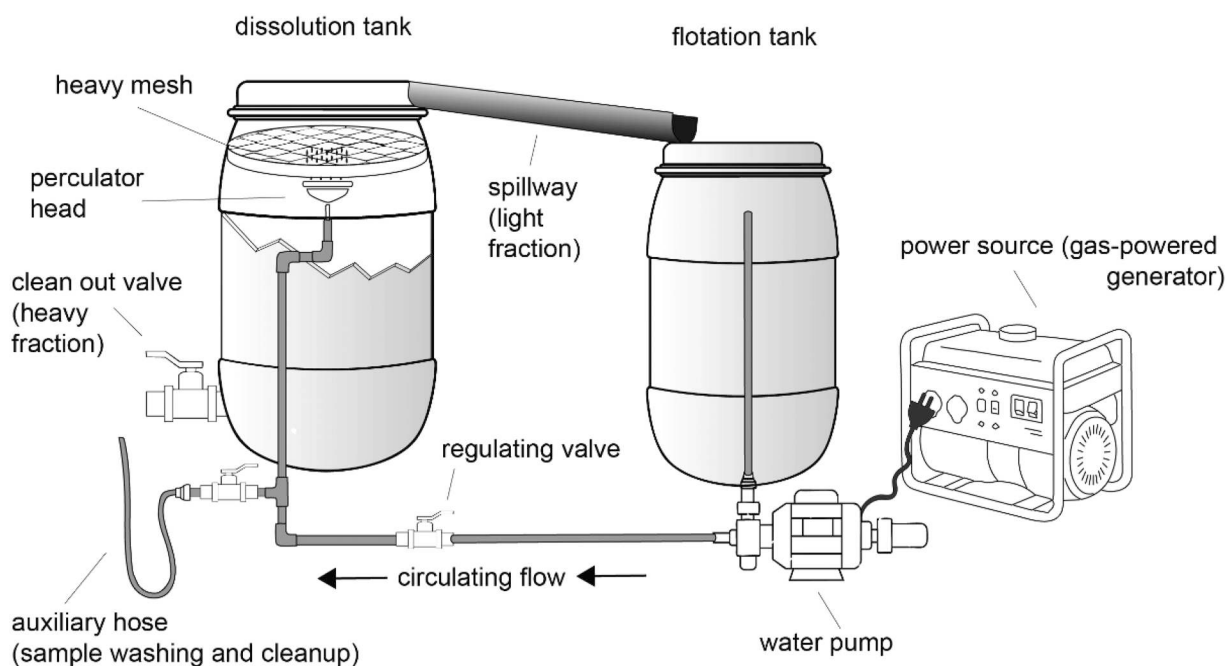


Figure 4. Diagram of the flotation system setup that was modified from the design in Hastorf (1999) to recover light and heavy fraction from adobe bricks.

as waste sediment. This process allows us to directly associate paleoecological, cultural, and radiocarbon materials when they are found within the same adobe packet.

After each sample was fully dissolved, the system cycled until the water ran clear of fine/light fraction. Screens were changed and washed between samples, and heavy fraction sediments and water within the tank were flushed to limit contamination. Floated light fraction samples were rinsed with DI H₂O and rapidly air dried in an enclosed lab to limit potential contamination from airborne particulates or new fungal growth.

¹⁴C analysis and modeling

We selected stratigraphically related ¹⁴C samples for radiocarbon analysis so that age estimates could be constrained by Bayesian models that incorporated sequential relationships as priors. We manually cleaned each AMS ¹⁴C sample using Milli-Q ultrapure water and a synthetic brush to remove foreign matter while examining them microscopically for rootlets, burrowing insect remains, or other intrusive sources of carbon. Cleaned samples underwent a pretreatment of successive acid-base-acid (ABA) washes at 70°C following procedures modified from the UC-Irvine KCCAMS facility protocols. Pretreated samples were washed again with Milli-Q to neutralize pH, capped and rapidly dried in a drying oven at 50–70°C, and then stored in a desiccation chamber until shipped for analysis. The accelerated mass spectrometry facilities at the Pennsylvania State University Institute of Energy and Environment Radiocarbon Laboratory (lab code PSUAMS), DirectAMS (lab code D-AMS), University of California-Irvine Keck Carbon Cycle Accelerator Mass Spectrometer Facility (lab code UCIAMS), or the Oxford Radiocarbon Accelerator Unit (lab code OxAU) analyzed samples. The resulting radiocarbon ages are provided in Table 2, while the reliable ages and relationships to contexts, cultural attributions, and adobe types used in the modeling are provided in Supplemental Material 4.

We calibrated all dates to the SHCal20 calibration curve (Hogg et al. 2020) in the R statistical environment using the **c14bazAAR** package (Schmid, Seidensticker, and Hinz 2019). Although selection of a calibration curve at low latitudes can be difficult, Chicama's subequatorial eastern Pacific location makes SHCal20 appropriate (Marsh et al. 2018). Where identifiable construction sequences make it possible to incorporate stratigraphic relationships as model priors, we calculated posterior estimates of sample ages using OxCal 4.4 (Ramsey 2001) and kernel density estimation (KDE) summaries of posteriors for recognizable adobe types by cross-referencing. OxCal code is available in Supplemental Material 5.

Chicama Valley environments are arid, and organic preservation is exceptional. While this results in abundant dateable material, it also presents potential complications. Three problems exist: old wood, protracted residence times, and material recycling. Each can produce mismatches between a dated event (incorporation of atmospheric carbon into organic tissue) and a target event (incorporation of a sample into a cultural context). The widely recognized old wood problem arises when samples come from long-lived woody species whose growth may significantly predate (by centuries) its actual harvest and subsequent cultural use. Residence

time problems occur when organic materials persist on a landscape long after the death of their source and before they become incorporated into cultural contexts. This is a particular concern in arid environments like Chicama, where organic tissues can be well-preserved. Recycling problems result when materials previously incorporated into cultural contexts are reincorporated into later ones. While adobe bricks themselves are difficult to move and reuse, the unfired bricks may be mined as raw material to fabricate new adobe, recycling older organics into the younger construction material.

The first two problems can be minimized by careful sample selection, avoiding wood and wood charcoal in favor of macrobotanical remains from short-lived species. Recycling is best detected by examining the consistency of dated samples against each other and chronological priors, bracketing dates from earlier and later events to identify reversals. We selected carbonized wood only when no other macrobotanical remains were present in a sample or to directly compare ages of carbonized wood to uncarbonized macrobotanicals to assess old wood and residence time effects. Samples were selected from stratigraphically related contexts so that age estimates could be constrained by Bayesian modeling that incorporates sequential relationships as priors.

Results and Discussion

Our approach to recovering radiocarbon and ecological proxies from adobe as sealed packets proved highly successful and generated new research questions. While disturbed contexts are often avoided, we found that they can be judiciously leveraged to collect reliable data. We achieved high success rates in recovering suitable AMS ¹⁴C material from adobe and other earthen construction materials. Importantly, these materials yielded reliable ages for target events. We observed no consistent offsets or widespread problems with the reliability of AMS ¹⁴C ages from adobe brick, as was reported by Dee and colleagues (2013). Calibrated ages are predominantly sequential, age reversals were rare and readily identifiable, and Bayesian models capture the timing of construction activities with precision and fidelity. Brick forms were identifiable either directly or from associated samples. Ecological materials, while not ubiquitous, were common, and changes in abundance were meaningful.

High recovery rate of organics/adobe

Our sampling strategy proved overall to be effective, rapid, and low cost. We documented 582 contexts in 39 exposures at 13 sites over the course of two brief field seasons. In total, this represents approximately 1400 person hours, during which we operated as a small crew of four to five people in a COVID "bubble" due to pandemic restrictions. Our strategy is not unique to the restrictions that initially made it necessary, however, and could readily be adapted to general survey techniques.

Disaggregating samples in our modified flotation system was easier and faster than we initially expected. The process took approximately 5–20 minutes depending on the size of the original sample. Flotation recovered macrobotanical samples weighing ≥ 0.01 g (the minimum approximate

mass recommended for AMS determinations on uncarbonized and carbonized organic material) from over three quarters ($n = 128$ or 77.4%) of the adobes we sampled (see Table 3). We opportunistically collected a smaller number of samples while documenting the exposures; due to the exceptional preservation of macrobotanical remains at many sites, these opportunities were abundant. This yielded a total of 192 samples suitable for radiocarbon analysis with minimal financial and labor investments. We further reduced costs by relying heavily on digital workflows to reduce in-field time allocated to documentation and establishing in-house pretreatment protocols. The coordinated digital and analog recording approach collected greater and more detailed information than possible with conventional recording methods, with the added benefit that we did not further disturb primary archaeological contexts.

Integrity and suitability for chronological refinement

We dated 72 samples from 31 profiles at all 13 sites (Figure 5). Of these, only nine samples produced ages that we consider unreliable based on their inconsistency with stratigraphic relationships (see Supplemental Material 4). Our approach was highly effective at producing the information that Bayesian age modeling requires. Bayesian approaches require prior information, including stratigraphic or other sequential relationships and assertions about temporal distributions of events. By taking advantage of large exposures in disturbed contexts, it was possible to document overall stratigraphic relationships, gather redundant data (multiple samples from single phases), and link radiocarbon ages to expected constructed sequences. Stratigraphic sequencing provided robust priors. These allowed us to evaluate the reliability of samples

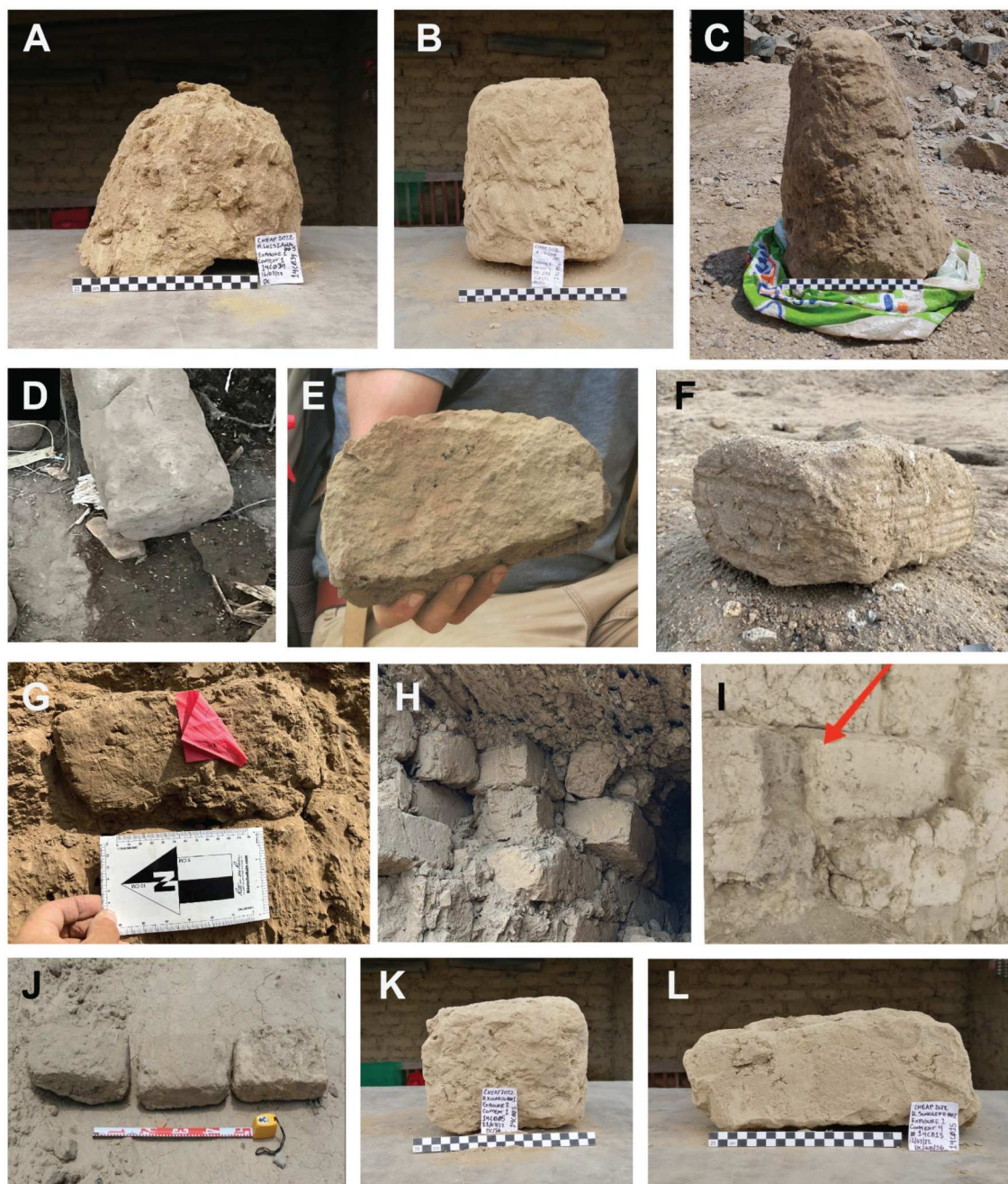


Figure 5. Examples of adobe forms from sites in the Chicama Valley. A) Short-conical or hemispherical (H. Luisiana), B) trunco-conical (H. Luisiana), C) large conical (H. Cruz la Botija), D) small conical (H. Pucuche), E) plano-convex (H. Facalá), F) rectangular cane-impressed (H. Santa Ana), G–H) two forms of rectangular smooth (H. Pampas de Jaguay), I) rectangular smooth (Mocollope), J) large rectangular (H. Sonolipe), K) large rectangular (H. Rosario), and, L) rectangular/irregular domed (H. Ongollape). Color versions of these figures are available online.

from adobe—material that is not typically regarded as a single depositional event—with close agreement between dated (sample age) and target (adobe manufacture) events.

We further tested the reliability of carbonized materials by pairing them with uncarbonized macrobotanical remains. Charcoal fragments are often regarded as less reliable, due to greater potential to preserve well and be recycled into later contexts. We have some indications that the reliability of materials extracted from adobe is context specific, especially where carbonized wood is involved. Large, well-preserved pieces of charcoal were abundant in conical adobes from Huacas Luisiana, Cruz la Botija, and Pucuchín. However, dates on this carbonized wood, ca. 2250–1925 B.C. (4000–3700 CAL B.P.), were highly consistent with each other, clustered tightly within a few centuries, produced no stratigraphic age reversals, and were consistent with dates on short-lived materials. Charcoal appears to have been much more abundant on the late Archaic Chicama landscape and is not remnant from earlier phases.

In contrast, charcoal was less abundant and highly fragmented in bricks from later periods (see below). Although unreliable ages were still rare, these more often yielded reversals in contexts younger than ca. A.D. 1150–1290. Companion dates on short-lived botanical materials from the same brick were consistent with stratigraphy, phasing, and expected dates, however. These apparent relationships need to be more systematically evaluated. Our experience indicates that problematic samples are less common than frequently assumed and can be avoided with careful sample selection. Problems of mismatches between dated events and target events can be minimized by dating uncarbonized remains, which are abundant and well preserved in younger Chicama sites.

Adobes as archives of ecological proxies

While we do not fully explore this potential here, we have provisional indications that adobes from Chicama offer a wealth of ecological information. Wood charcoal abundance and size vary greatly by site (see Table 3), suggesting broader landscape and ecological changes could be reconstructed. Wood charcoal comprised 0.001–0.21% weight of the adobe sampled from Huacas Luisiana, Pucuchín, and Cruz la Botija (ca. 2200 B.C.). Charcoal in the highest yielding adobes from Luisiana approximates an estimated 59.5 cm³, or about 21% of the entire brick by volume. In contrast, charcoal is much less abundant and highly fragmented in later sites, such as Huacas Rosario and Sonolipe (ca. A.D. 1100–1400). The variable abundances and type of subfossil organic material preserved in mudbrick offer interesting lines of inquiry into the timing and pace of ecological change in Chicama in tandem with early monumental construction. This evidence supports models suggesting early global shifts toward a “used planet,” in which human land use drove rapid, widespread ecological change and established largely anthropogenic ecological networks (e.g., Ellis et al. 2013).

Adobes/mudbrick may present exceptional opportunities to reconstruct paleoecology. Arbøll and colleagues (2023) found botanical aDNA from 34 identifiable taxonomic groups within unfired mudbrick from Nimrud (883–859 B.C.). Patalano and colleagues (2022) reconstructed environmental conditions from plant (*Fragmites*) wax *n*-alkanes and carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) isotopes in Han Dynasty

(ca. 132 and 116 B.C.) segments of the Great Wall. Microbotanical remains—pollen, phytoliths, and starch grains—are abundant in Huaca de la Luna adobe; these vary between structures and stratigraphic units, hinting at changes in environments where the materials were sourced (Alvaro Quispe 2020). The utility of adobes as environmental archives stems from the fact that ecological proxies and radiometrically datable material are encased within effectively sealed sedimentary packets and in their greater context can be related directly to the cultural milieu of monumental construction.

Adobes as chronologically-sensitive markers

We test assumptions of chronological sensitivity of adobe bricks by modeling calibrated ¹⁴C age KDEs for identifiable types. Our results show that previously proposed relative sequences hold up to scrutiny, broadly speaking; however, several assumptions regarding the age of certain forms, sequentiality versus contemporaneity of types, and their chronological sensitivity are not supported (see Figures 5–8). We found direct equivalencies between the types we sampled and those previously described by Reindel, Gálvez Mora, and others. As Reindel’s typology is the most comprehensive, we relied on this as a “Rosetta Stone” to link types we encountered to other systems (see Figure 8).

Certain types are chronologically well-constrained (see Figures 5–8). Large conical adobes (*n* = 19 from eight construction episodes at three sites) fall within an approximately 300 year period. Modeled within stratigraphic sequences, these ages produce posterior estimates summarized by a KDE with a 95% high density region (HDR) between 2245 and 1925 B.C. (see Figures 5A–C, 6). The consistency in ages across these three sites is remarkable and surprising. Moreover, it suggests that there was a burst of monument construction throughout Chicama at the end of the 3rd millennium B.C. that, among other things, used very similar architectural canons.

Mold-made forms are distinctive types whose chronology and cultural affiliation have been debated. One mold type is made from river cane (*Phragmites comunis*) or palm petioles bound together (Alvaro Quispe 2020), leaving concave impressions on the adobe (see Figure 5F). Another type of mold was constructed from wooden planks, resulting in smooth-sided bricks (see Figure 5G). Reindel (1993) places cane-impressed bricks, of three subtypes at 18 sites, into the Early Intermediate Period generally without ascribing a more explicit age range or cultural affiliation. Chappelaine, Pimentel, and Gamboa (2009), citing various authors, states that cane-marked adobe bricks “have traditionally been considered a Gallinazo-specific feature;” some sources they cite, however, also link this type to the Virú culture (Bennett 1950; Strong and Evans 1952; Wilson 1988). Hastings and Moseley (1975) argue distinct types of molds are associated with Virú, Gallinazo, and Moche constructions. Our sample of rectangular cane-impressed adobes (*n* = 6, from six construction episodes at three sites) falls within the span A.D. 205–745 (summary of posterior estimates from models that incorporate stratigraphic priors, KDE 95% HDR; see Figure 6).

Rectangular smooth-sided adobes (*n* = 7, from seven construction episodes at three sites) produce a bimodal KDE (see Figures 5, 6). Rectangular smooth adobes (see Figure 5G–I)

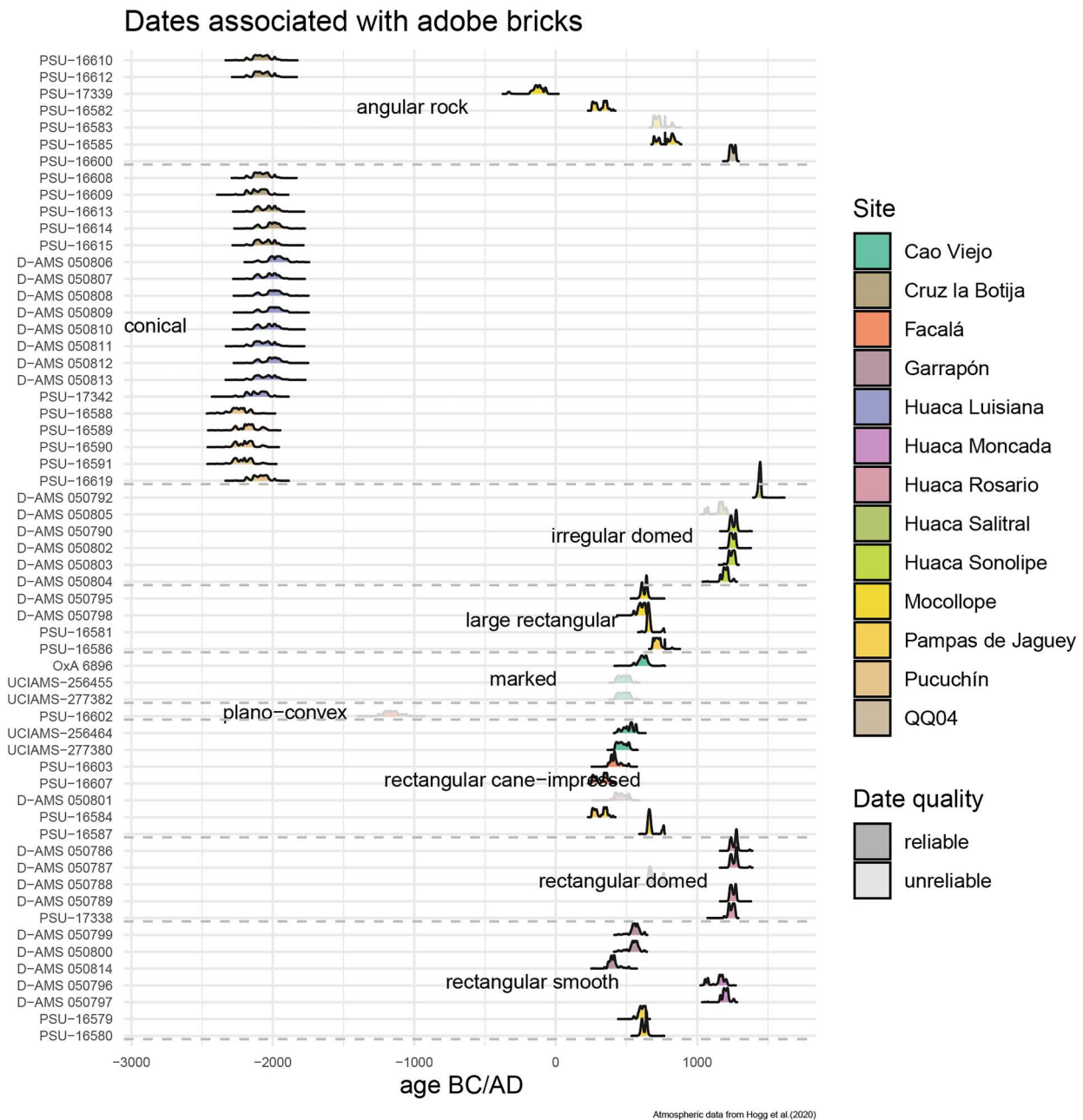


Figure 6. Calibrated radiocarbon ages from adobe bricks and related construction materials arranged according to brick and material type. Dates are calibrated to the SHCal20 curve using atmospheric data from Hogg and colleagues (2020). Color versions of these figures are available online.

from Huaca Santa Ana and Pampas de Jagüey fall in the span of A.D. 260–840 CE. Similar adobes from Huaca Moncada fall in the span of A.D. 985–1365 (summaries of posterior estimates from models that incorporate stratigraphic priors, KDE 95% HDR). The bimodal distribution of ages suggests that different successive cultural groups used bricks that were similar in form and manufacture. Because of the basic characteristics and simplicity of smooth-sided, rectangular brick types, this form may have less chronological specificity or separability relative to other types. Other types—large rectangular, rectangular domed, and irregular domed—are distinguishable by their size class and the form of their upper surface (planar or convex, see Figure 5J–L). Despite these variations in form, radiometric dates and KDEs of posteriors for these types overlap with each other and with other parallelepiped forms (see Figures 6, 7).

The earlier distribution of ages for rectangular smooth-sided adobes partially overlaps with the cane-impressed

adobe model (see Figures 7, 8). Distinctive impressions on the sides of well-preserved cane-impressed and smooth-sided adobes make it unlikely that the latter are simply eroded vestiges of the former. We suggest two manufacturing techniques were being used contemporaneously, rather than sequentially, during the 3rd–8th centuries A.D. to produce two distinct adobe types. Previous authors (Bazán and Alva 2024; Gálvez Mora et al. 2003; Hastings and Moseley 1975) noted similar overlap as earlier cane-impressed forms were gradually replaced by rectangular smooth ones. Our radiometric ages indicate that this period of contemporaneous use was longer and there was not necessarily sequential replacement. Both rectangular smooth and cane-impressed adobes also overlap chronologically with what we term large rectangular adobes (see Figures 7, 8). These are distinguishable from both cane-impressed and rectangular smooth types by their greater height (ca. 15 cm rather than 8–12 cm). Large rectangular adobes ($n = 4$, from four construction episodes

KDEs of posteriors associated with adobe bricks

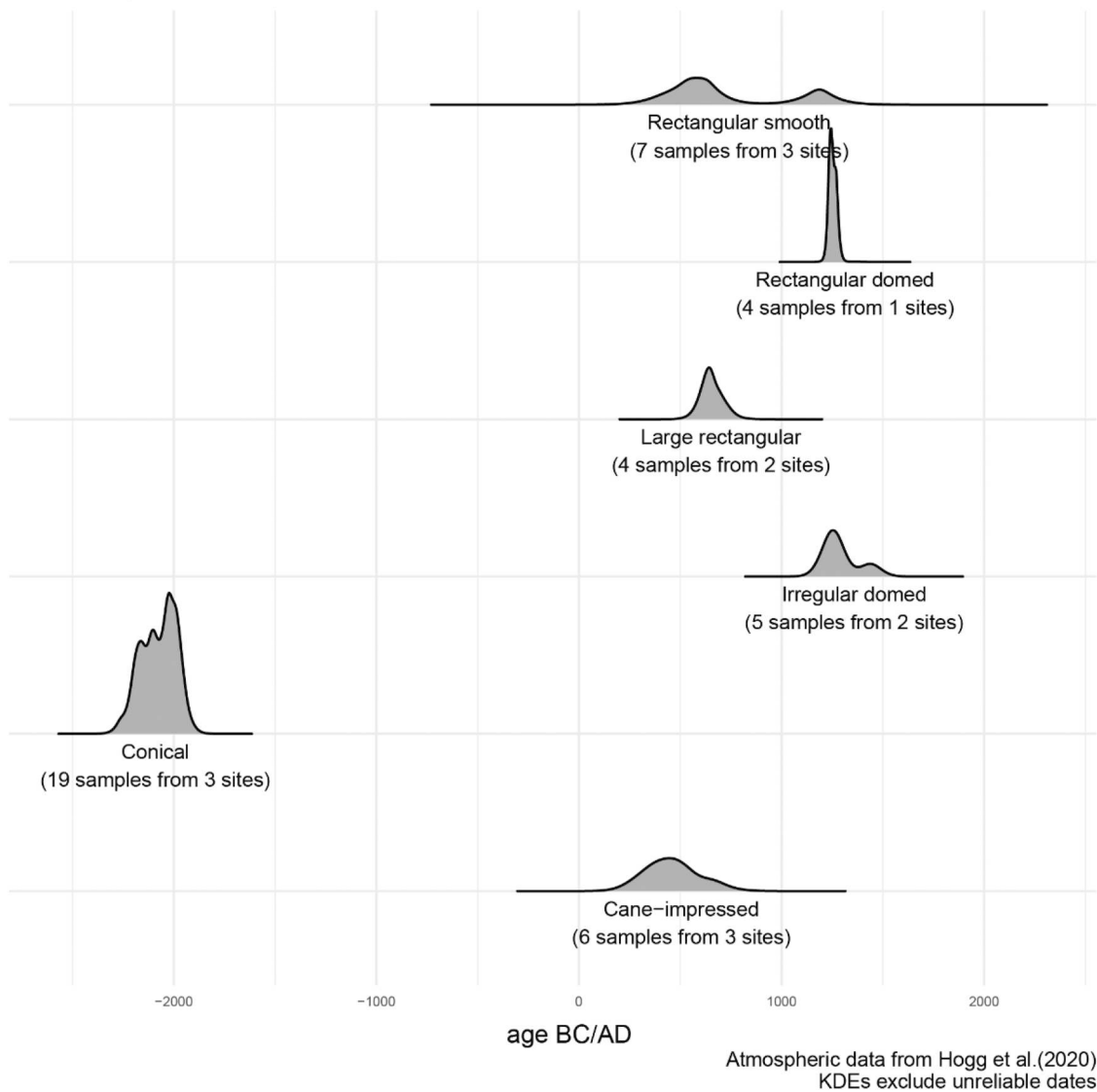


Figure 7. Kernel density estimates (KDEs) of posteriors for adobe brick types (note: the plano-convex type is not represented in this plot, as the date on this type was unreliable).

at two sites) fall within the span of A.D. 545–775 (summary of posterior estimates from models that incorporate stratigraphic priors, KDE 95% HDR).

Reindel (1993, table 5) separated adobes that are rectangular in plan with domed tops into three sequential types, differentiated by the height of the top (“low domed,” “normal domed,” and “highly domed;” see Figure 7). We separate similar adobes into only two categories to avoid subjectively classifying types based on the degree of roundedness of the top. These are rectangular domed and irregular domed; the latter is distinguished by its apparently casual manufacture and rough finish (see Figures 5, 7). Rectangular domed adobes ($n = 4$, from four construction episodes at one site) fall within the span of A.D. 1215–1290 CE (summary of posterior estimates from models that incorporate stratigraphic priors, KDE 95% HDR). Irregular domed adobes ($n = 5$, from four construction episodes at two sites) fall within the span of A.D. 1145–1500 (summary of posterior estimates from models that incorporate stratigraphic priors, KDE 95% HDR).

Stone masonry construction, finally, may have little chronological specificity (see Figure 6). Constructions of

angular stone, though less common than adobe, are found in various Chicama sites. Associated dates ($n = 6$, from six construction episodes at four sites) span approximately 2100 B.C.–A.D. 800 (see Figure 5). Use of angular stone, we infer, is not reliably a chronological feature and may relate to other factors such as material availability.

In summary, adobe forms provide chronological information, but their interpretation can be complicated. Figure 8 compares the posterior KDEs resulting from our radiometric dating program to schematic chronological and typological sequences for the Chicama Valley (and other adjacent valleys). The KDEs, which describe the age distributions of identified types, illustrate more complex temporal relationships than Reindel’s and Gálvez Mora’s sequential or Campana Delgado’s phylogenetic models argue. Large conical forms are the only ones that are clearly isolated in time, apparently providing an unambiguous and very early temporal marker. Our results suggest a longer timespan for the cane-impressed type than was previously proposed. The bimodal distribution of rectangular smooth forms prevents ascribing that type to a single period, while portions overlap with distributions for rectangular domed and irregular

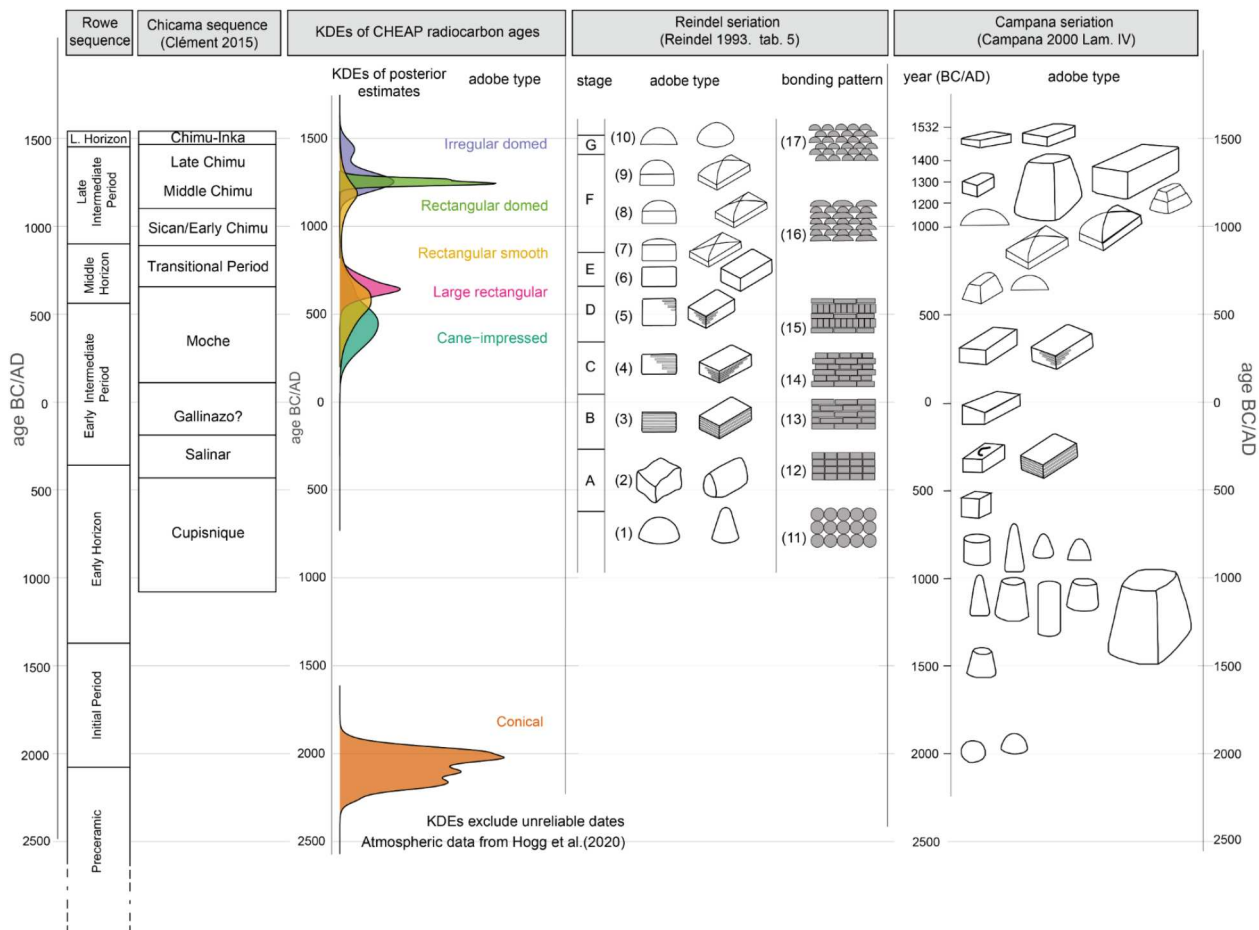


Figure 8. Comparison of the radiometric kernel density estimates (KDEs) of posteriors for adobe brick types compared with previously published brick typologies based on seriations. The general chronological sequence used is from Clément (2015), which modifies the Larco sequence. Chicama Archaeological Project (CHEAP) KDEs are modeled using the SHCAL20 calibration and atmospheric data from Hogg and colleagues (2020). Chronological scales redrawn from Reindel (1993) and Campana Delgado (2000) are approximate. For the Reindel typology, the brick types and bonding patterns are: 1) conical/hemispherical, 2) miscellaneous hand-formed, 3) flat with cane impressions, 4) flat with smoothed cane impressions, 5) flat with scoring marks, 6) flat smoothed, 7) low domed, 8) normal or moderately domed, 9) highly domed, 10) planoconvex, 11) without or stacked bond, 12) stacked bond, 13) running bond, 14) English bond, 15) rolling bond or *cabeza y soga* (header and stretcher), 16) running or English bond, and, 17) running bond. Color versions of these figures are available online.

domed types. Posterior age estimates associated with cane-impressed, some rectangular smooth, and large rectangular types also overlap. Multiple types were in use concurrently, even while the advent of some forms is sequentially earlier than others. Complexities in the radiometric models highlight some limitations of using adobe forms as chronological and cultural markers, as well as the importance of considering multiple lines of chronological evidence in assessing construction episodes.

New insights into construction pacing from radiometric ages

Directly dating adobe bricks enables us to identify and confront three fundamental limitations of existing chronological schema: 1) an inability to connect type seriations (adobe or otherwise) to absolute, calendrical dates; 2) the coarse resolution that results from assigning types to a position in a sequence that is poorly constrained chronologically; and, 3) assumptions that types are necessarily sequential.

The large conical adobe samples we dated directly at late Archaic sites illustrate these problems. Examples of this type we sampled proved to be considerably earlier than previously proposed but consistent with the relative sequence of adobe forms. This underscores a key challenge in chronology

building: achieving accuracy without sacrificing precision. While prior schema accurately placed conical adobes early in sequences, they did so by being quite vague about when exactly they were in use. Reindel's typology suggests an age range of over 1000 years for this type. Similarly, conical adobes excavated by Bird at Huaca Prieta are bracketed by associated TPQ/TAQ dates approximately 900 years apart (Bird, Hyslop, and Skinner 1985, 49–50, tables 1, 3, fig. 31). In contrast, our radiometric ages cluster strongly within an approximately 270 year period and place this type at the very end of the Late Archaic, ca. 2245–1925 B.C., at least a few centuries earlier than the Formative Period (Cupisnique) affiliation previously proposed (Toshihara 2002; Ubbelohde-Doering 1952). The improved precision and accuracy of radiometric ages demonstrate that this type was used at the sites we sampled during a relatively brief explosion of monument construction in the Late Archaic rather than belonging to a protracted, millennium-long architectural tradition. This requires us to reconsider not only what period we believe conical adobes to indicate but also how we envision cultural changes at the end of the Late Archaic.

One of the weaknesses of seriation is its assumption of sequence, which radiocarbon ages reveal can be optimistic. As discussed above, radiometric dating indicates greater complexity in the sequencing of various cane-impressed, rectangular, or parallelepiped forms than was previously

proposed. Some of the precision that was inherent in assigning forms to certain cultural affiliations may not be supported. The original publications of adobe typologies often emphasized that the frequencies of types vary, and one form gradually replaces another. In practice, this nuance is often glossed over in favor of the tacit assumption that types were used sequentially. The overlap we document demonstrates that this tacit assumption is unwarranted.

Adobe types do not appear to reflect a phenomenon of one uniform type replacing a similarly well-defined predecessor in simplistic sequences. Rather, there is more apparent complexity and variability in types that were used contemporaneously during any one time, especially after approximately A.D. 400. These results caution us to closely consider the limitations of lower resolution, lower precision, and potentially inaccurate relative chronologies.

Perhaps most importantly, these results further suggest that much regarding building choices, including adobe selection, remains to be explored; why builders chose one construction type over another likely reflects, at any given moment, a constellation of material, technical, and socio-behavioral decisions. The widespread and contemporaneous use of a common type across the Chicama Valley—such as we have documented in large conical adobes—may be indicative of an expansive social tradition in the late Archaic that maintained communities of practice through monument building. Conversely, the partially-overlapping use of varied rectangular, parallelepiped, and similar forms in the late 1st millennium A.D. may indicate greater heterogeneity—social or otherwise—was involved in monument construction. Recent reanalysis of Moche chronology similarly finds greater heterogeneity in contemporaneous ceramic types than was previously recognized, suggesting “heretofore-unknown social factors may have been at play” (Koons et al. 2024, 93).

A punctuated model of monumental construction

The chronology we developed from these 72 ^{14}C AMS dates suggests that monumental huaca construction was not evenly distributed over time. Rather, monument building appears to have proliferated during relatively brief periods of heightened activity, with longer construction hiatuses in between (see Figures 7, 8). Given concerns about biases affecting the set of previously published Chicama radiocarbon dates (see Figures 1, 2), we selected sampling locations to capture a more complete transect across the valley. Moreover, we targeted sites which, estimating from adobe forms and associated diagnostic ceramics, would capture a more comprehensive temporal sequence. Despite this strategy, all reliable dates we produced ($n = 72$) fall within three temporal spans, between approximately 2200 and 1900 B.C., A.D. 200 and 850, and A.D. 1100 and 1500.

The median ages for 19 dates from Luisana, Cruz la Botija, and Pucuchín fall within 270 years of each other but are separated from the next dated cultural phase by a > 1200 year hiatus. This cluster of dates is surprising for several reasons: the ages are highly consistent, they belong to a much earlier period at the end of the late Archaic than previously estimated, and these dates derive from eight distinct construction events—both early and late in each site’s sequence—at three very large monumental complexes across the lower and middle valley. Our results suggest there was an explosion in

monumental construction activity during the span 2200–1900 B.C. in the Chicama Valley. The second and third clusters of dates between ca. A.D. 250 and 1500 are notable, as they similarly suggest that intensive construction was concentrated within relatively brief periods, separated by hiatuses of relatively very little activity (see Figure 7). Collectively, these dates suggest that monumental complexes were often constructed contemporaneously and rapidly within certain periods, rather than accreting over much longer periods of time and/or built sequentially one after the other.

We propose that the construction of monuments (but not necessarily their use) in Chicama may reflect a distinctly punctuated rather than a continuous or gradualist practice. The construction phases accessible through these exposures cluster within discrete, century-scale periods, rather than extending continuously across the ca. 3700 years that our sample covers. The result is an ebb and flow pattern (sensu Marsh et al. 2025) defined by periods of intensified construction that were separated by longer cessations in construction. This pattern is consistent with recent studies of well-dated North Coast contexts, which document brief but intensive construction campaigns that lasted as little as several decades but were separated by hiatuses of several centuries (Bazán 2022; e.g., Millaire 2020, 8; Sghinolfi, Millaire, and Roy 2023). These concentrated episodes of activity are short-lived relative to the valley’s millennia-long history and the cultural periods to which those sites are conventionally assigned. Future dates could, of course, change this hypothesis; however, the current pattern of strongly clustered dates in our sample, together with the previously published Chicama radiocarbon record and evidence from adjacent valleys, is compelling.

A tacit assumption of continuous activity is a legacy of temporal periodization schemes. Monuments built through gradual, incremental, or evenly paced construction would be expected to yield radiocarbon dates reflecting a consistent tempo. Instead, the strongly uneven distribution of dates reported here supports a pattern of brief pulses of intensified monument construction. Even if these apparent hiatuses are filled by new samples at additional sites, that would imply sequential rather than simultaneous huaca construction in the Chicama Valley, with major implications for population size estimates and interpretations of sociopolitical organization.

Conclusion

The approach we discuss above allows archaeologists to capitalize on the unfortunate reality of pervasive site disturbances in order to sample contexts on the North Coast of Peru that otherwise are accessible only through conventional and more costly intensive excavations. Sampling construction sequences exposed by looting and other disturbances is logistically straightforward. It provides an effective means for rapidly assessing multiple sites; documenting the timing, sequencing, and pace of construction sequences; refining radiometric chronologies; and, ultimately, addressing critical anthropological questions.

Dating adobe, in our experience, was overwhelmingly successful: these construction materials, while humble and often overlooked, are ubiquitous and information-rich. We had high rates of recovery of organic materials, and we encountered fewer issues of old carbon, residence time,

and recycling than is often assumed. The stratigraphic sequences exposed by looting were conducive to combining radiocarbon ages with priors about construction sequences, significantly improving our control of the resulting chronologies. Even if we are not dating all construction episodes, this approach allows us to compare dates of construction activity across sites.

Chronology and sequencing have always been central to understanding the diachronic dimension of cultural change. Nonetheless, the conceptual frameworks and diagrams that were developed as methodological tools often reinforce, implicitly or explicitly, notions of relatively constant cultural evolution (Ramón Joffré 2005). We found broad support for the relative sequence of adobe typologies that have been previously published; however, we also found strong indications that the specific chronological assumptions of these sequences were inaccurate. Several types were earlier than previously assumed; radiocarbon ages for nearly all types indicate that they were used during much more abbreviated periods and belonged to chronologically shorter building traditions than previously assumed; and, several subvarieties or subtypes appear to have been used concurrently rather than sequentially. That is, the development and use of these types was more complex than a simple linear sequence. Our radiometric findings are consistent with a recent, very robust reanalysis of Moche radiocarbon ages (Koons et al. 2024), which proposed that the Moche culture was a much briefer, more intensive, and socially heterogeneous cultural development than was previously understood.

Furthermore, by directly dating adobe and other construction materials, we found that the frequencies of monument building were often inconstant. These results show that radiometric ages of adobe and similar construction materials can be used to examine the temporal patterns (timing and tempo) of cultural activities within the homogeneous block-like cultural periodization schema that we have inherited from prior investigations. On the basis of our initial results, we hypothesize that monumental construction activities across Chicama Valley were very uneven in time, occurring in very discrete pulses of activity with protracted hiatuses in between. Radiometric ages not only allow us to recognize this pattern but also offer a means to further analyze its significance. It would be interesting to try to link this pacing to endogenous or exogenous factors that might explain the fluctuations in activity. This analysis results in several new hypotheses and questions concerning diachronic patterns of monumental construction that warrant further testing.

The particular archaeological circumstances of the Peruvian North Coast are exceptional, but by no means are they unique. Monumental adobe architecture that has experienced some degree of destruction is commonplace throughout Andean South America and has parallels in other arid and semi-arid regions of the world where earthen architecture was used. This philosophical and methodological approach could readily be applied to those regions; furthermore, we suspect that similar approaches could be used to develop positive outcomes out of the unfortunate reality of sites' partial destruction worldwide.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

- Alvaro Quispe, L. P. 2020. "Saberes Tecnológicos Moche Materializados Durante El Proceso de Producción de Los Adobes Del Templo Viejo y Templo Nuevo Del Complejo Arqueológico Huacas Del Sol y de La Luna." Tesis de maestría, Universidad Nacional de Trujillo.
- Arbøll, T. P., S. L. Rasmussen, N. de Jonge, A. H. Hansen, C. Pertoldi, and J. L. Nielsen. 2023. "Revealing the Secrets of a 2900-Year-Old Clay Brick, Discovering a Time Capsule of Ancient DNA." *Scientific Reports* 13 (1): 13092. doi:10.1038/s41598-023-38191-w.
- Attarian, C. J. 2003. "Pre-Hispanic Urbanism and Community Expression in the Chicama Valley, Peru." Ph.D. Dissertation, University of California, Los Angeles.
- Attarian, C. J. 2009. "Urbanism and Social Change During the Gallinazo and Moche Periods in the Chicama Valley." In *Gallinazo: An Early Cultural Tradition on the Peruvian North Coast*, edited by J.-F. Millaire, and M. Morlion, 77–90. Los Angeles: Cotsen Institute of Archaeology Press.
- Bazán, A. 2022. "Datación Absoluta de La Secuencia Constructiva de La Huaca Cao Viejo, Valle de Chicama." In *Cronologías de La Costa Norte Del Peru: Recientes Estudios y Panoramas*. Trujillo: VIII Coloquio Internacional de Arqueología FW-CAEB.
- Bazán, A. 2023. Apuntes acerca de la ocupación Lambayeque en el valle de Chicama y en el complejo arqueológico El Brujo. En R. Buitron, *Fardos Lambayeque en El Brujo. Catálogo de la Colección*, 15–54. Lima: Fundación Augusto N. Wiese.
- Bazán, A., and J. Alva, eds. 2024. *Reevaluando la cronología de la Huaca Cao Viejo, Complejo Arqueológico El Brujo. Reporte integral de investigaciones de campo de la temporada 2020*. Lima: Fundación Augusto N. Wiese.
- Bennett, W. C. 1950. *The Gallinazo Group, Virú Valley, Peru*. Publications in Anthropology 43. New Haven, CT: Yale University.

- Bird, J. B. 1948. "Pre-ceramic Cultures in Chicama and Viru." *Memoirs of the Society for American Archaeology* 4:21–28. doi:10.1017/S0081130000000320.
- Bird, J. B., J. Hyslop, and M. D. Skinner. 1985. *The Preceramic Excavations at the Huaca Prieta Chicama Valley, Peru*. Vol. 62. no. 1. New York: American Museum of Natural History NY.
- Brooks, W. E., J. C. Willett, J. D. Kent, V. Vasquez, and T. Rosales. 2005. "The Muralla Pircada—An Ancient Andean Debris Flow Retention Dam, Santa Rita B Archaeological Site, Chao Valley, Northern Peru." *Landslides* 2:117–23. doi:10.1007/s10346-005-0051-7.
- Campana Delgado, C. 2000. *Tecnologías Constructivas De Tierra En La Costa Norte Prehispánica*. Lima: Insituto Nacional de Cultura.
- Carmichael, P. H. 2019. "Stages, Periods, Epochs, and Phases in Paracas and Nasca Chronology: Another Look at John Rowe's Ica Valley Master Sequence." *Nawpa Pacha* 39 (2): 145–79. doi:10.1080/00776297.2019.1623468.
- Castillo, L. J., and C. B. Donnan. 1994. "Los Mochicas Del Norte y Los Mochicas Del Sur." In *Viciús*, edited by K. Makowski, C. B. Donnan, and I. A. Bullon, 143–181. Lima: Banco de Crédito del Perú.
- Cavallaro, R., and I. Shimada. 1988. "Some Thoughts on Sican Marked Adobes and Labor Organization." *American Antiquity* 53 (1): 75–101. doi:10.2307/281155.
- Chapdelaine, C. 2011. "Recent Advances in Moche Archaeology." *Journal of Archaeological Research* 19 (2): 191–231. doi:10.1007/s10814-010-9046-6.
- Chapdelaine, C., V. Pimentel, and J. Gamboa. 2009. "Gallinazo Cultural Identity in the Lower Santa Valley: Ceramics, Architecture, Burial Patterns, and Sociopolitical Organization." In *Gallinazo: An Early Cultural Tradition on the Peruvian North Coast*, edited by J.-F. Millaire and M. Morlion, 181–206. Los Angeles: UCLA Cotsen Institute of Archaeology Press.
- Chauchat, C. 1975. "The Paján Complex, Pampa de Cupisnique, Peru." *Nawpa Pacha: Journal of Andean Archaeology* : 85–96. doi:10.1179/naw.1975.13.1.006.
- Cleland, K. M., and I. Shimada. 1992. "Sican Bottles: Marking Time in the Peruvian Bronze Age—A Five-Part Typology and Seriation." *Andean Past* 3 (15): 193–235.
- Clément, C. 2015. *Paysage Socioculturel et Architecture Dans La Culture Chimú: L'implantation Humaine à L'intermédiaire Recent (1000-1470 Apr. J.-C.) Dans La Vallée de Chicama (Côte Nord Du Pérou)*. Oxford: BAR International Series/Archaeopress.
- Contreras, D. A., E. J. Marsh, and K. Rademaker. 2025. "Leveraging Radiocarbon in the Central Andes: From Chronologies to Research Agendas." *Quaternary International* 746:109957. doi:10.1016/j.quaint.2025.109957.
- Dee, M., D. Wengrow, A. Shortland, A. Stevenson, F. Brock, L. G. Flink, and C. B. Ramsey. 2013. "An Absolute Chronology for Early Egypt Using Radiocarbon Dating and Bayesian Statistical Modelling." *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 469 (2159): 20130395. doi:10.1098/rspa.2013.0395.
- Dillehay, T. D. 2017. *Where the Land Meets the Sea: Fourteen Millennia of Human History at Huaca Prieta, Peru*. Austin, TX: University of Texas Press.
- Dillehay, T. D., D. Bonavia, S. Goodbred, M. Pino, and V. Vasquez. 2012. "Chronology, Mound-Building and Environment at Huaca Prieta, Coastal Peru, from 13,700 to 4000 Years Ago." *Antiquity* 86 (331), doi:10.1017/S0003598X00062451.
- Donnan, C. B. 1973. "Moche Occupation of the Santa Valley." Ph.D. dissertation, University of California, Berkeley.
- Elera Arévalo, C. G. 1998. "The Puémapé Site and the Cupisnique Culture: A Case Study on the Origins and Development of Complex Society in the Central Andes, Perú." Ph.D. Dissertation, University of Calgary.
- Ellis, E. C., J. O. Kaplan, D. Q. Fuller, S. Vavrus, K. K. Goldewijk, and P. H. Verburg. 2013. "Used Planet: A Global History." *Proceedings of the National Academy of Sciences* 110 (20): 7978–85. doi:10.1073/pnas.1217241110.
- Feathers, J., J. Quilter, and S. LeBlanc. 2024. "Luminescence Dating of Adobe from Monumental Architecture in North Coastal Peru." *Journal of Archaeological Science: Reports* 56:104568. doi:10.1016/j.jasrep.2024.104568.
- Franco Jordán, R. 2016. *Mocollope, Pasado Prehispánico*. Chiclayo, Peru: Imprenta Ingenium.
- Franco Jordán, R., C. A. Gálvez Mora, and S. Vásquez Sánchez. 2003. "Modelos, Función y Cronología de La Huaca Cao Viejo, Complejo El Brujo." In *Moche: hacia el final del milenio. Actas del Segundo Coloquio sobre la Cultura Moche*, edited by Santiago Uceda y Elías Mujica, Trujillo. Lima: Universidad Nacional de Trujillo y Pontificia Universidad Católica del Perú.
- Fuchs, P. R., R. Patzschke, G. Yenque, and J. Briceño. 2009. "Del Arcaico Tardío al Formativo Temprano: Las Investigaciones En Sechín Bajo, Valle de Casma." *Boletín de Arqueología PUCP* 13:55–86. doi:10.18800/boletindearqueologiapucp.200901.002.
- Gamboa, J., and George, L., eds. 2022. *Paisaje, identidad y memoria: la sociedad Recuay (100-800 dC) y los Andes Norcentrales de Peru*. Lima: Staatliche Museen zu Berlin.
- Gálvez Mora, C. 2012. "Adobe, Tiempo y Arquitectura En El Valle de Chicama: 1300 aC–1100 dC." *C. Arkinka* 16:88–97.
- Gálvez Mora, C. A., R. Becerra, and R. Marín. 2002. *Inventario de Sitios Arqueológicos de La Provincia de Ascope. Distrito de Chicama, Santiago de Cao y Magdalena de Cao (Primera Parte)*. Lima, Peru: Report submitted to the Instituto Nacional de Cultura.
- Gálvez, C., A. Murga, D. Vargas, and H. Ríos. 2003. "Secuencia y Cambios En Los Materiales y Técnicas Constructivas de La Huaca Cao Viejo, Complejo El Brujo, Valle de Chicama." In *Moche: Hacia el Final del Milenio, Actas del Segundo Coloquio sobre la Cultura Moche*, edited by S. Uceda and E. Mujica, 79–118. Lima: Universidad Nacional de Trujillo y Pontificia Universidad Católica del Perú.
- Gálvez Mora, C., and J. Briceño. 2001. "The Moche in the Chicama Valley." In *Moche Art and Archaeology in Ancient Peru*, vol. 63, edited by J. Pilsbury, 141–158. New Haven and London: Yale University Press.
- Gálvez Mora, C., and J. C. Murga. 2014. "Arquitectura Post-Mochica Elaborada En Tierra: La Evidencia Del Valle de Chicama." in *Cultura Lambayeque en el contexto de la costa norte del Perú. Actas del primer y segundo coloquio*, edited by J. C. Fernández Alvarado and C. E. Wester La Torre, 397–418. Chiclayo: Emdecosege S.A.
- Goldberg, A., A. M. Mychajliw, and E. A. Hadly. 2016. "Post-Invasion Demography of Prehistoric Humans in South America." *Nature* 532 (7598): 232–35. doi:10.1038/nature17176.
- Hastings, C. M., and M. E. Moseley. 1975. "The Adobes of Huaca Del Sol and Huaca de La Luna." *American Antiquity* 40 (2Part1): 196–203. doi:10.2307/279615.
- Hasstorf, C. A. 1999. "A Manual for Building a Mechanized Flotation Machine, Modified from the SMAP Machine First Described by Patty Jo Watson in 1976." *UC Berkeley McCown Archaeobotany Laboratory Reports* (42).
- Hogg, A. G., T. J. Heaton, Q. Hua, J. G. Palmer, C. S. Turney, J. Southon, A. Bayliss, P. G. Blackwell, G. Boswijk, C. B. Ramsey, C. Pearson, F. Petchey, P. Reimer, R. Reimer, and L. Wacker. 2020. "SHCal20 Southern Hemisphere Calibration, 0–55,000 Years Cal BP." *Radiocarbon* 62 (4): 759–78. doi:10.1017/RDC.2020.59.
- Huckleberry, G., A. Caramanica, and J. Quilter. 2017. "Dating the Ascope Canal System: Competition for Water During the Late Intermediate Period in the Chicama Valley, North Coast of Peru." *Journal of Field Archaeology* : 1–14. doi:10.1080/00934690.2017.1384662.
- Kolata, A. L. 1978. *Chan Chan: The Form of the City in Time*. [Ph.D. dissertation]. Cambridge, MA: Harvard University.
- Kolata, A. L. 1982. "Chronology and Settlement Growth at Chan Chan." In *Chan Chan: Andean Desert City*, edited by Michael M. Moseley and Kent C. Day, 67–85. Albuquerque: University of New Mexico Press.
- Koons, M. L. 2015. "Moche Sociopolitical Dynamics and the Role of Licapa II, Chicama Valley, Peru." *Latin American Antiquity* 26 (4): 473–92. doi:10.7183/1045-6635.26.4.473.
- Koons, M. L., and B. A. Alex. 2014. "Revised Moche Chronology Based on Bayesian Models of Reliable Radiocarbon Dates." *Radiocarbon* 56 (3): 1039–55. doi:10.2458/56.16919.
- Koons, M. L., B. C. Rizzuto, L. Trever, A. Boswell, A. B. Perez, L. A. Muro Ynoán, G. Prieto, C. Rengifo, K. Sharp, E. Swenson, H. Ikehara-Tsukayama, J. Ortiz Zevallos, T. Cotrina Roncal, R. J. George, J. M. Capriles, and F. Tokanai. 2024. "Moche Chronology of Ancient Peru: Bayesian Assessment of Radiocarbon Dates and Ceramic Styles from North to South." *Quaternary International* 703: 82–96. <https://doi.org/https://doi.org/10.1016/j.quaint.2024.05.008>.
- Kosok, P. 1965. *Life, Land and Water in Ancient Peru*. Brooklyn, New York: Long Island University Press.

- Kus, J. S. 1972. *Selected Aspects of Irrigated Agriculture in the Chimú Heartland, Peru* [Ph.D. dissertation]. University of California, Los Angeles.
- Larco Hoyle, R. 1938. *Los Mochicas*. Lima: Museo Arqueológico Rafael Larco Herrera.
- Larco Hoyle, R. 1945. *Los Cupisniques*. Buenos Aires: Sociedad Geográfica Americana.
- Larco Hoyle, R. 1946. "A Culture Sequence for the North Coast of Perú." In *Handbook of South American Indians, Vol. 2*, edited by J. H. Steward. Washington, DC: Smithsonian Institution, Bureau of American Ethnology.
- Larco Hoyle, R. 1948. *Cronología Arqueológica Del Norte Del Perú*. Buenos Aires: Sociedad Geográfica Americana.
- Leonard, B., and G. Russell. 1992. "Informe Preliminar: Proyecto de Reconocimiento Arqueológico Del Chicama, Resultados de La Primera Temporada de Campo, 1989." Lima: Report Submitted to the Instituto Nacional de Cultura.
- Marsh, E. J., M. C. Bruno, S. C. Fritz, P. Baker, J. M. Capriles, and C. A. Hastorf. 2018. "IntCal, SHCal, or a Mixed Curve? Choosing a 14 C Calibration Curve for Archaeological and Paleoenvironmental Records from Tropical South America." *Radiocarbon* : 1–16. doi:10.1017/RDC.2018.16.
- Marsh, E. J., B. Owen, A. Korpisaari, N. Sharratt, P. Goldstein, B. Vining, S. Baitzel, S. deFrance, M. Hubbe, and S. Ancapichún. 2025. "Dating the Ebb and Flow of Tiwanaku and Post-Collapse Material Culture Across the Andes." *Quaternary International* 727:109742. doi:10.1016/j.quaint.2025.109742.
- Mauricio, A. C., R. Grieseler, A. R. Heller, A. R. Kelley, F. Rumiche, D. H. Sandweiss, and W. Viveen. 2021. "The Earliest Adobe Monumental Architecture in the Americas." *Proceedings of the National Academy of Sciences* 118 (48): e2102941118.
- Millaire, J.-F. 2009. "Gallinazo and the Tradición Norcosteña." In *Gallinazo: An Early Cultural Tradition on the Peruvian North Coast*, edited by J.-F. Millaire y M. Morlion, 1–16. Los Angeles: UCLA Cotsen Institute of Archaeology Press.
- Millaire, J.-F. 2020. "Dating the Occupation of Cerro Arena: A Defensive Salinar-Phase Settlement in the Moche Valley, Peru." *Journal of Anthropological Archaeology* 57:101142.
- Millaire, J.-F., and M. Morlion, eds. 2009. *Gallinazo: An Early Cultural Tradition on the Peruvian North Coast*. Los Angeles: UCLA Cotsen Institute of Archaeology Press.
- Millaire, J.-F., G. Prieto, F. Surette, E. M. Redmond, and C. S. Spencer. 2016. "Statecraft and Expansionary Dynamics: A Virú Outpost at Huaca Prieta, Chicama Valley, Peru." *Proceedings of the National Academy of Sciences* 113 (41): E6016–25.
- Moseley, M. E. 1975. "Prehistoric Principles of Labor Organization in the Moche Valley, Peru." *American Antiquity* 40 (2Part1): 191–96. doi:10.1073/pnas.1609972113.
- Moubarac, J.-C. 2002. *La Production D'adobes Dans L'ancienne Capitale de L'état Moche, Société Précolombienne de La Côte Nord Du Pérou*. doi:10.2307/279614.
- Narvaez, A. 1989. "Chan Chan: Chronology and Stratigraphic Contents." *Andean Past* 2 (1): 9.
- Oakland, A. 1999. *Informe preliminar de análisis de fardos funerarios de Huaca Cao Viejo*. Unpublished report, Complejo Arqueológico El Brujo. Magdalena de Cao.
- Pardo, C., and J. Rucabado, eds. 2016. *Moche y Sus Vecinos: Reconstruyendo Identidades*. Lima: Museo de Arte de Lima.
- Patalano, R., J. Hu, Q. Leng, W. Liu, H. Wang, P. Roberts, M. Storzum, L. Yang, and H. Yang. 2022. "Ancient Great Wall Building Materials Reveal Environmental Changes Associated with Oases in Northwestern China." *Scientific Reports* 12 (1): 22517.
- Pozorski, T. G. 1987. "Changing Priorities Within the Chimú State the Role of Irrigation Agriculture." In *The Origins and Development of Andean State*, edited by J. Haas, S. Pozorski, and T. Pozorski, 111–120. Cambridge: Cambridge University Press.
- Quilter, J. 2011. "Cultural Encounters at Magdalena de Cao Viejo in the Early Colonial Period." In *Enduring Conquests: Rethinking the Archaeology of Resistance to Spanish Colonialism in the Americas*, edited by M. Liebmann, and M. S. Murphy, 103–26. Santa Fe, NM: School of Advanced Research Press.
- Ramón Joffré, G. 2005. "Periodificación En Arqueología Peruana: Genealogía y Aporía." *Bulletin de l'Institut Français d'études Andines* 34 (1): 5–33.
- Ramsey, C. B. 2001. "Development of the Radiocarbon Calibration Program." *Radiocarbon* 43 (2A): 355–63. doi:10.4000/bifea.5567.
- Reimer, P. J., T. A. Brown, and R. W. Reimer. 2004. "Discussion: Reporting and Calibration of Post-Bomb 14C Data." *Radiocarbon* 46 (3): 1299–304. doi:10.1017/S0033822200038212.
- Reindel, M. 1993. *Baumaterialien, Konstruktionstechniken Und Bauformen Der Monumentalen Lehmarchitektur an Der Nordküste Perus*. Mainz: Von Zabern. doi:10.1017/S0033822200033154.
- Reindel, M. 1997. "Aproximación a La Arquitectura Monumental de Adobe En La Costa Norte Del Perú." In *Archaeologica Peruana 2. Architecture and Civilization in the Prehispanic Andes*, edited by E. Bonnier and H. Bischof, 91–106. Heidelberg: Sociedad Arqueológica Peruano-Alemana/ReissMuseum Mannheim.
- Roddick, A. P. 2016. "Scalar Relations: A Juxtaposition of Craft Learning in the Lake Titicaca Basin." In *Knowledge in Motion: Constellations of Learning Across Time and Place*. Amerind Studies in Archaeology. University of Arizona Press Tucson.
- Russell, G. S., B. L. Leonard, J. R. Briceño, and J. Briceño. 1994. "Cerro Mayal: Nuevos Datos Sobre La Producción de Cerámica Moche En El Valle de Chicama." In *Moche: propuestas y perspectivas*, edited by S. Uceda and E. Mujica, 181–206. Lima: Universidad Nacional de Trujillo-Institut Français d'Études Andines.
- Santana-Sagredo, F., E. Dufour, N. Goepfert, A. Zazzo, R. F. Jordán, and S. V. Sánchez. 2020. "New Bioarchaeological Evidence and Radiocarbon Dates from the Lambayeque/Sicán Culture Camelids from the El Brujo Complex (Northern Coast of Peru): Implications for Funerary and Herd Management Practices." *Environmental Archaeology* 25 (3): 333–52.
- Schmid, C., D. Seidensticker, and M. Hinz. 2019. "c14bazAAR: An R Package for Downloading and Preparing C14 Dates from Different Source Databases." *Journal of Open Source Software* 4 (43): 1914. doi:10.1080/14614103.2018.1556960.
- Sghinolfi, A., J.-F. Millaire, and A. Roy. 2023. "Feeding the Desert: Radiocarbon Dating the Salinar Phase in the Virú Valley, Peruvian North Coast." *Journal of Archaeological Science: Reports* 49:104031. doi:10.21105/joss.01914.
- Shimada, I. 1990. "Cultural Continuities and Discontinuities on the Northern North Coast of Peru, Middle-Late Horizons." In *The Northern Dynasties: Kingship and Statecraft in Chimor*, edited by M. Moseley and A. Cordy-Collins, 297–392. Washington, DC: Dumbarton Oaks.
- Shimada, I. 1994. *Pampa Grande and the Mochica Culture*. Austin, TX: University of Texas Press.
- Shimada, I., and R. Cavallaro. 1985. "Monumental Adobe Architecture of the Late Prehispanic Northern North Coast of Peru." *Journal de La Société Des Américanistes* : 41–78.
- Shimada, I., and A. Maguina. 1994. "Nueva Visión Sobre La Cultura Gallinazo y Su Relación Con La Cultura Moche." In *Moche: Propuestas y Perspectivas*, edited by S. Uceda and E. Mujica, 31–58. Lima: Universidad Nacional de Trujillo-Institut Français d'Études Andines.
- Strong, W. D., and C. Evans. 1952. *Cultural Stratigraphy in the Virú Valley Northern Peru: The Formative and Florescent Epochs*. New York: Columbia University Press.
- Stuiver, M., and H. A. Polach. 1977. "Discussion: Reporting of 14C Data." *Radiocarbon* 19 (3): 355–63.
- Tavera-Medina, C., H. Tantaleán, C. Stanish, J. Quispe-Baquedano, D. Huachaca, J. Roman, and A. Boswell. 2025. "Quebrada Del Oso: A Chimú Agricultural Production Centre in the Chicama Valley, North Coast of Perú." *Antiquity* : 1–8. doi:10.1017/S0033822200003672.
- Toshihara, K. 2002. *The Cupisnique Culture in the Formative Period World of the Central Andes, Peru* [Ph.d. Dissertation]. University of Illinois at Urbana-Champaign. doi:10.15184/aqy.2025.10261.
- Tsai, H. I. 2012. "Adobe Bricks and Labor Organization on the North Coast of Peru." *Andean Past* 10 (1): 9.
- Tschauner, H. 2001. *Socioeconomic and Political Organization in the Late Prehispanic Lambayeque Sphere, Northern North Coast of Peru* [Ph.d. Dissertation]. Harvard University.

- Ubbelohde-Doering, H. 1952. *The Art of Ancient Peru*. New York: Frederick A. Praeger.
- Uceda, S., H. Gayoso, F. Castillo, and C. Rengifo. 2021. "Climate and Social Changes: Reviewing the Equation with Data from the Huacas de Moche Archaeological Complex, Peru." *Latin American Antiquity* 32 (4): 705–22.
- Unkel, I., B. Kromer, M. Reindel, L. Wacker, and G. Wagner. 2007. "A Chronology of the Pre-Columbian Paracas and Nasca Cultures in South Peru Based on AMS 14 C Dating." *Radiocarbon* 49 (2): 551–64. doi:10.1017/laq.2021.35.
- Vining, B. R. 2017. "Cultural Niche Construction and Remote Sensing of Ancient Anthropogenic Environmental Change in the North Coast of Peru." *Journal of Archaeological Method and Theory* : 1–28. doi:10.1017/S0033822200042466.
- Wenger, E. 1999. *Communities of Practice: Learning, Meaning, and Identity*. Cambridge: Cambridge University Press. doi:10.1007/s10816-017-9346-y
- Wilson, D. J. 1988. *Prehispanic Settlement Patterns in the Lower Santa Valley, Peru: A Regional Perspective on the Origins and Development of Complex North Coast Society*. Washington, DC: Smithsonian Institution Press.